3.1. Cross-validation: evaluating estimator performance
3.2. Tuning the hyper-parameters of an estimator
3.3. Tuning the decision threshold for class prediction
3.4. Metrics and scoring: quantifying the quality of predictions
3.5. Validation curves: plotting scores to evaluate models
4. Metadata Routing
5. Inspection
6. Visualizations
7. Callbacks
8. Dataset transformations
9. Dataset loading utilities
10. Computing with scikit-learn
11. Model persistence
12. Common pitfalls and recommended practices
13. Data Interoperability
14. Choosing the right estimator
15. External Resources, Videos and Talks

User Guide > 3. Model selection and evaluation > 3.4. Metrics and scoring: quantifying the quality of predictions

On this page
3.4.1. Which scoring function should I use?
3.4.2. Scoring API overview
3.4.3. The `scoring` parameter: defining model evaluation rules
3.4.3.1. String name scorers
3.4.3.2. Callable scorers
3.4.3.3. Using multiple metric evaluation
3.4.4. Classification metrics
3.4.5. Multilabel ranking metrics
3.4.6. Regression metrics
3.4.7. Clustering metrics
3.4.8. Dummy estimators

scikit-learn is financially supported by Probabl and other institutions.

:probabl.
Enterprise-grade solutions and services

3.4. Metrics and scoring: quantifying the quality of predictions

3.4.1. Which scoring function should I use?

Before we take a closer look into the details of the many scores and evaluation metrics, we want to give some guidance, inspired by statistical decision theory, on the choice of **scoring functions** for **supervised learning**, see [Gneiting2009]:

- *Which scoring function should I use?*
- *Which scoring function is a good one for my task?*

In a nutshell, if the scoring function is given, e.g. in a kaggle competition or in a business context, use that one. If you are free to choose, it starts by considering the ultimate goal and application of the prediction. It is useful to distinguish two steps:

- Predicting
- Decision making

Predicting: Usually, the response variable Y is a random variable, in the sense that there is *no deterministic* function $Y = g(X)$ of the features X. Instead, there is a probability distribution F of Y. One can aim to predict the whole distribution, known as *probabilistic prediction*, or—more the focus of scikit-learn—issue a *point prediction* (or point forecast) by choosing a property or functional of that distribution F. Typical examples are the mean (expected value), the median or a quantile of the response variable Y (conditionally on X).

Once that is settled, use a **strictly consistent** scoring function for that (target) functional, see [Gneiting2009]. This means using a scoring function that is aligned with *measuring the distance between predictions* `y_pred` *and the true target functional using observations of* Y, i.e. `y_true`. For classification **strictly proper scoring rules**, see Wikipedia entry for Scoring rule and [Gneiting2007], coincide with strictly consistent scoring functions. The table further below provides examples. One could say that consistent scoring functions act as *truth serum* in that they guarantee *"that truth telling […] is an optimal strategy in expectation"* [Gneiting2014].

Once a strictly consistent scoring function is chosen, it is best used for both: as loss function for model training and as metric/score in model evaluation and model comparison.

Note that for regressors, the prediction is done with predict while for classifiers it is usually predict_proba.

Decision Making: The most common decisions are done on binary classification tasks, where the result of predict_proba is turned into a single outcome, e.g., from the predicted probability of rain a decision is made on how to act (whether to take mitigating measures like an umbrella or not). For classifiers, this is what predict returns. See also Tuning the decision threshold for class prediction. There are many scoring functions which measure different aspects of such a decision, most of them are covered with or derived from the **`metrics.confusion_matrix`**.

List of strictly consistent scoring functions: Here, we list some of the most relevant statistical functionals and corresponding strictly consistent scoring functions for tasks in practice. Note that the list is not complete and that there are more of them. For further criteria on how to select a specific one, see [Fissler2022].

functional	scoring or loss function	response `y`	prediction
Classification			
mean	Brier score [1]	multi-class	`predict_proba`
mean	log loss	multi-class	`predict_proba`
mode	zero-one loss [2]	multi-class	`predict`, categorical
Regression			
mean	squared error [3]	all reals	`predict`, all reals
mean	Poisson deviance	non-negative	`predict`, strictly positive
mean	Gamma deviance	strictly positive	`predict`, strictly positive
mean	Tweedie deviance	depends on `power`	`predict`, depends on `power`
median	absolute error	all reals	`predict`, all reals
quantile	pinball loss	all reals	`predict`, all reals
mode	no consistent one exists	reals	

[1] The Brier score is just a different name for the squared error in case of classification with one-hot encoded targets.

[2] The zero-one loss is only consistent but not strictly consistent for the mode. The zero-one loss is equivalent to one minus the accuracy score, meaning it gives different score values but the same ranking.

[3] R^2 gives the same ranking as squared error.

Fictitious Example: Let's make the above arguments more tangible. Consider a setting in network reliability engineering, such as maintaining stable internet or Wi-Fi connections. As provider of the network, you have access to the dataset of log entries of network connections containing network load over time and many interesting features. Your goal is to improve the reliability of the connections. In fact, you promise your customers that on at least 99% of all days there are no connection discontinuities larger than 1 minute. Therefore, you are interested in a prediction of the 99% quantile (of longest connection interruption duration per day) in order to know in advance when to add more bandwidth and thereby satisfy your customers. So the *target functional* is the 99% quantile. From the table above, you choose the pinball loss as scoring function (fair enough, not much choice given), for model training (e.g. `HistGradientBoostingRegressor(loss="quantile", quantile=0.99)`) as well as model evaluation (`mean_pinball_loss(..., alpha=0.99)` - we apologize for the different argument names, `quantile` and `alpha`) be it in grid search for finding hyperparameters or in comparing to other models like `QuantileRegressor(quantile=0.99)`.

References

[Gneiting2007] T. Gneiting and A. E. Raftery. Strictly Proper Scoring Rules, Prediction, and Estimation In: Journal of the American Statistical Association 102 (2007), pp. 359– 378. link to pdf

[Gneiting2009](1,2) T. Gneiting. Making and Evaluating Point Forecasts Journal of the American Statistical Association 106 (2009): 746 - 762.

[Gneiting2014] T. Gneiting and M. Katzfuss. Probabilistic Forecasting. In: Annual Review of Statistics and Its Application 1.1 (2014), pp. 125–151.

[Fissler2022] T. Fissler, C. Lorentzen and M. Mayer. Model Comparison and Calibration Assessment: User Guide for Consistent Scoring Functions in Machine Learning and Actuarial Practice.

3.4.2. Scoring API overview

There are 3 different APIs for evaluating the quality of a model's predictions:

- **Estimator score method**: Estimators have a `score` method providing a default evaluation criterion for the problem they are designed to solve. Most commonly this is accuracy for classifiers and the coefficient of determination (R^2) for regressors. Details for each estimator can be found in its documentation.
- **Scoring parameter**: Model-evaluation tools that use cross-validation (such as **`model_selection.GridSearchCV`**, **`model_selection.validation_curve`** and **`linear_model.LogisticRegressionCV`**) rely on an internal *scoring* strategy. This can be specified using the `scoring` parameter of that tool and is discussed in the section The scoring parameter: defining model evaluation rules.
- **Metric functions**: The **`sklearn.metrics`** module implements functions assessing prediction error for specific purposes. These metrics are detailed in sections on Classification metrics, Multilabel ranking metrics, Regression metrics and Clustering metrics.

Finally, Dummy estimators are useful to get a baseline value of those metrics for random predictions.

See also

For "pairwise" metrics, between *samples* and not estimators or predictions, see the Pairwise metrics, Affinities and Kernels section.

3.4.3. The `scoring` parameter: defining model evaluation rules

Model selection and evaluation tools that internally use cross-validation (such as **`model_selection.GridSearchCV`**, **`model_selection.validation_curve`** and **`linear_model.LogisticRegressionCV`**) take a `scoring` parameter that controls what metric they apply to the estimators evaluated.

They can be specified in several ways:

- `None`: the estimator's default evaluation criterion (i.e., the metric used in the estimator's `score` method) is used.
- String name: common metrics can be passed via a string name.
- Callable: more complex metrics can be passed via a custom metric callable (e.g., function).

Some tools do also accept multiple metric evaluation. See Using multiple metric evaluation for details.

3.4.3.1. String name scorers

For the most common use cases, you can designate a scorer object with the `scoring` parameter via a string name; the table below shows all possible values. All scorer objects follow the convention that **higher return values are better than lower return values**. Thus metrics which measure the distance between the model and the data, like **`metrics.mean_squared_error`**, are available as 'neg_mean_squared_error' which return the negated value of the metric.

Scoring string name	Function	Comment
Classification		
'accuracy'	`metrics.accuracy_score`	
'balanced_accuracy'	`metrics.balanced_accuracy_score`	
'top_k_accuracy'	`metrics.top_k_accuracy_score`	
'average_precision'	`metrics.average_precision_score`	
'neg_brier_score'	`metrics.brier_score_loss`	requires `predict_p` support
'f1'	`metrics.f1_score`	for binary targets
'f1_micro'	`metrics.f1_score`	micro-avera
'f1_macro'	`metrics.f1_score`	macro-aver
'f1_weighted'	`metrics.f1_score`	weighted average
'f1_samples'	`metrics.f1_score`	by multilabe sample
'neg_log_loss'	`metrics.log_loss`	requires `predict_p` support
'precision' etc.	`metrics.precision_score`	suffixes app as with 'f1'
'recall' etc.	`metrics.recall_score`	suffixes app as with 'f1'
'jaccard' etc.	`metrics.jaccard_score`	suffixes app as with 'f1'
'roc_auc'	`metrics.roc_auc_score`	
'roc_auc_ovr'	`metrics.roc_auc_score`	
'roc_auc_ovo'	`metrics.roc_auc_score`	
'roc_auc_ovr_weighted'	`metrics.roc_auc_score`	
'roc_auc_ovo_weighted'	`metrics.roc_auc_score`	
'd2_log_loss_score'	`metrics.d2_log_loss_score`	requires `predict_p` support
'd2_brier_score'	`metrics.d2_brier_score`	requires `predict_p` support
Clustering		
'adjusted_mutual_info_score'	`metrics.adjusted_mutual_info_score`	
'adjusted_rand_score'	`metrics.adjusted_rand_score`	
'completeness_score'	`metrics.completeness_score`	
'fowlkes_mallows_score'	`metrics.fowlkes_mallows_score`	
'homogeneity_score'	`metrics.homogeneity_score`	
'mutual_info_score'	`metrics.mutual_info_score`	
'normalized_mutual_info_score'	`metrics.normalized_mutual_info_score`	
'rand_score'	`metrics.rand_score`	
'v_measure_score'	`metrics.v_measure_score`	
Regression		
'explained_variance'	`metrics.explained_variance_score`	
'neg_max_error'	`metrics.max_error`	
'neg_mean_absolute_error'	`metrics.mean_absolute_error`	
'neg_mean_squared_error'	`metrics.mean_squared_error`	
'neg_root_mean_squared_error'	`metrics.root_mean_squared_error`	
'neg_mean_squared_log_error'	`metrics.mean_squared_log_error`	
'neg_root_mean_squared_log_error'	`metrics.root_mean_squared_log_error`	
'neg_median_absolute_error'	`metrics.median_absolute_error`	
'r2'	`metrics.r2_score`	
'neg_mean_poisson_deviance'	`metrics.mean_poisson_deviance`	
'neg_mean_gamma_deviance'	`metrics.mean_gamma_deviance`	
'neg_mean_absolute_percentage_error'	`metrics.mean_absolute_percentage_error`	
'd2_absolute_error_score'	`metrics.d2_absolute_error_score`	

Usage examples:

```
>>> from sklearn import svm, datasets
>>> from sklearn.model_selection import cross_val_score
>>> X, y = datasets.load_iris(return_X_y=True)
>>> clf = svm.SVC(random_state=0)
>>> cross_val_score(clf, X, y, cv=5, scoring='recall_macro')
array([0.96, 0.96, 0.96, 0.93, 1.  ])
```

Note

If a wrong scoring name is passed, an `InvalidParameterError` is raised. You can retrieve the names of all available scorers by calling **`get_scorer_names`**.

3.4.3.2. Callable scorers

For more complex use cases and more flexibility, you can pass a callable to the `scoring` parameter. This can be done by:

- Adapting predefined metrics via make_scorer
- Creating a custom scorer object (most flexible)

3.4.3.2.1. Adapting predefined metrics via `make_scorer`

The following metric functions are not implemented as named scorers, sometimes because they require additional parameters, such as **`fbeta_score`**. They cannot be passed to the `scoring` parameters; instead their callable needs to be passed to **`make_scorer`** together with the value of the user-settable parameters.

Function	Parameter	Example usage
Classification		
`metrics.fbeta_score`	`beta`	`make_scorer(fbeta_score, beta=2)`
Regression		
`metrics.mean_tweedie_deviance`	`power`	`make_scorer(mean_tweedie_deviance, power=1.5)`
`metrics.mean_pinball_loss`	`alpha`	`make_scorer(mean_pinball_loss, alpha=0.95)`
`metrics.d2_tweedie_score`	`power`	`make_scorer(d2_tweedie_score, power=1.5)`
`metrics.d2_pinball_score`	`alpha`	`make_scorer(d2_pinball_score, alpha=0.95)`

One typical use case is to wrap an existing metric function from the library with non-default values for its parameters, such as the `beta` parameter for the **`fbeta_score`** function:

```
>>> from sklearn.metrics import fbeta_score, make_scorer
>>> ftwo_scorer = make_scorer(fbeta_score, beta=2)
>>> from sklearn.model_selection import GridSearchCV
>>> from sklearn.svm import LinearSVC
>>> grid = GridSearchCV(LinearSVC(), param_grid={'C': [1, 10]},
...                     scoring=ftwo_scorer, cv=5)
```

The module **`sklearn.metrics`** also exposes a set of simple functions measuring a prediction error given ground truth and prediction:

- functions ending with `_score` return a value to maximize, the higher the better.
- functions ending with `_error`, `_loss`, or `_deviance` return a value to minimize, the lower the better. When converting into a scorer object using **`make_scorer`**, set the `greater_is_better` parameter to `False` (`True` by default; see the parameter description below).

3.4.3.2.2. Creating a custom scorer object

You can create your own custom scorer object using **`make_scorer`**.

Custom scorer objects using `make_scorer`

Using custom scorers in functions where n_jobs > 1

3.4.3.3. Using multiple metric evaluation

Scikit-learn also permits evaluation of multiple metrics in `GridSearchCV`, `RandomizedSearchCV` and `cross_validate`.

There are three ways to specify multiple scoring metrics for the `scoring` parameter:

- As an iterable of string metrics:

  ```
  >>> scoring = ['accuracy', 'precision']
  ```

- As a `dict` mapping the scorer name to the scoring function:

  ```
  >>> from sklearn.metrics import accuracy_score
  >>> from sklearn.metrics import make_scorer
  >>> scoring = {'accuracy': make_scorer(accuracy_score),
  ...            'prec': 'precision'}
  ```

  Note that the dict values can either be scorer functions or one of the predefined metric strings.

- As a callable that returns a dictionary of scores:

  ```
  >>> from sklearn.model_selection import cross_validate
  >>> from sklearn.metrics import confusion_matrix
  >>> # A sample toy binary classification dataset
  >>> X, y = datasets.make_classification(n_classes=2, random_state=0)
  >>> svm = LinearSVC(random_state=0)
  >>> def confusion_matrix_scorer(clf, X, y):
  ...      y_pred = clf.predict(X)
  ...      cm = confusion_matrix(y, y_pred)
  ...      return {'tn': cm[0, 0], 'fp': cm[0, 1],
  ...              'fn': cm[1, 0], 'tp': cm[1, 1]}
  >>> cv_results = cross_validate(svm, X, y, cv=5,
  ...                             scoring=confusion_matrix_scorer)
  >>> # Getting the test set true positive scores
  >>> print(cv_results['test_tp'])
  [10  9  8  7  8]
  >>> # Getting the test set false negative scores
  >>> print(cv_results['test_fn'])
  [0 1 2 3 2]
  ```

3.4.4. Classification metrics

The **`sklearn.metrics`** module implements several loss, score, and utility functions to measure classification performance. Some metrics might require probability estimates of the positive class or non-thresholded decision values (as returned by decision_function on some classifiers). Most implementations allow each sample to provide a weighted contribution to the overall score, through the `sample_weight` parameter.

Some of these are restricted to the binary classification case:

`precision_recall_curve` (y_true, y_score, *[, ...])	Compute precision-recall pairs for different probability thresholds.
`roc_curve` (y_true, y_score, *[, pos_label, ...])	Compute Receiver operating characteristic (ROC).
`class_likelihood_ratios` (y_true, y_pred, *[, ...])	Compute binary classification positive and negative likelihood ratios.
`det_curve` (y_true, y_score[, pos_label, ...])	Compute Detection Error Tradeoff (DET) for different probability thresholds.
`confusion_matrix_at_thresholds` (y_true, ...)	Calculate binary confusion matrix terms per classification threshold.

Others also work in the multiclass case:

`balanced_accuracy_score` (y_true, y_pred, *[, ...])	Compute the balanced accuracy.
`cohen_kappa_score` (y1, y2, *[, labels, ...])	Compute Cohen's kappa: a statistic that measures inter-annotator agreement.
`confusion_matrix` (y_true, y_pred, *[, ...])	Compute confusion matrix to evaluate the accuracy of a classification.
`hinge_loss` (y_true, pred_decision, *[, ...])	Average hinge loss (non-regularized).
`matthews_corrcoef` (y_true, y_pred, *[, ...])	Compute the Matthews correlation coefficient (MCC).
`roc_auc_score` (y_true, y_score, *[, average, ...])	Compute Area Under the Receiver Operating Characteristic Curve (ROC AUC) from prediction scores.
`top_k_accuracy_score` (y_true, y_score, *[, ...])	Top-k Accuracy classification score.

Some also work in the multilabel case:

`accuracy_score` (y_true, y_pred, *[, ...])	Accuracy classification score.
`classification_report` (y_true, y_pred, *[, ...])	Build a text report showing the main classification metrics.
`f1_score` (y_true, y_pred, *[, labels, ...])	Compute the F1 score, also known as balanced F-score or F-measure.
`fbeta_score` (y_true, y_pred, *, beta[, ...])	Compute the F-beta score.
`hamming_loss` (y_true, y_pred, *[, sample_weight])	Compute the average Hamming loss.
`jaccard_score` (y_true, y_pred, *[, labels, ...])	Jaccard similarity coefficient score.
`log_loss` (y_true[, y_proba, normalize, ...])	Log loss, aka logistic loss or cross-entropy loss.
`multilabel_confusion_matrix` (y_true, y_pred, *)	Compute a confusion matrix for each class or sample.
`precision_recall_fscore_support` (y_true, ...)	Compute precision, recall, F-measure and support for each class.
`precision_score` (y_true, y_pred, *[, labels, ...])	Compute the precision.
`recall_score` (y_true, y_pred, *[, labels, ...])	Compute the recall.
`roc_auc_score` (y_true, y_score, *[, average, ...])	Compute Area Under the Receiver Operating Characteristic Curve (ROC AUC) from prediction scores.
`zero_one_loss` (y_true, y_pred, *[, ...])	Zero-one classification loss.
`d2_log_loss_score` (y_true[, y_proba, ...])	D^2 score function, fraction of log loss explained.

And some work with binary and multilabel (but not multiclass) problems:

`average_precision_score` (y_true, y_score, *)	Compute average precision (AP) from prediction scores.

In the following sub-sections, we will describe each of those functions, preceded by some notes on common API and metric definition.

3.4.4.1. From binary to multiclass and multilabel

Some metrics are essentially defined for binary classification tasks (e.g. **`f1_score`**, **`roc_auc_score`**). In these cases, by default only the positive label is evaluated, assuming by default that the positive class is labelled `1` (though this may be configurable through the `pos_label` parameter).

In extending a binary metric to multiclass or multilabel problems, the data is treated as a collection of binary problems, one for each class. There are then a number of ways to average binary metric calculations across the set of classes, each of which may be useful in some scenario. Where available, you should select among these using the `average` parameter.

- `"macro"` simply calculates the mean of the binary metrics, giving equal weight to each class. In problems where infrequent classes are nonetheless important, macro-averaging may be a means of highlighting their performance. On the other hand, the assumption that all classes are equally important is often untrue, such that macro-averaging will over-emphasize the typically low performance on an infrequent class.
- `"weighted"` accounts for class imbalance by computing the average of binary metrics in which each class's score is weighted by its presence in the true data sample.
- `"micro"` gives each sample-class pair an equal contribution to the overall metric (except as a result of sample-weight). Rather than summing the metric per class, this sums the dividends and divisors that make up the per-class metrics to calculate an overall quotient. Micro-averaging may be preferred in multilabel settings, including multiclass classification where a majority class is to be ignored.
- `"samples"` applies only to multilabel problems. It does not calculate a per-class measure, instead calculating the metric over the true and predicted classes for each sample in the evaluation data, and returning their (`sample_weight`-weighted) average.
- Selecting `average=None` will return an array with the score for each class.

While multiclass data is provided to the metric, like binary targets, as an array of class labels, multilabel data is specified as an indicator matrix, in which cell `[i, j]` has value 1 if sample `i` has label `j` and value 0 otherwise.

3.4.4.2. Accuracy score

The **`accuracy_score`** function computes the accuracy, either the fraction (default) or the count (normalize=False) of correct predictions.

In multilabel classification, the function returns the subset accuracy. If the entire set of predicted labels for a sample strictly match with the true set of labels, then the subset accuracy is 1.0; otherwise it is 0.0.

If $\hat{y}_i$ is the predicted value of the i-th sample and y_i is the corresponding true value, then the fraction of correct predictions over n_samples is defined as

$$\texttt{accuracy}(y, \hat{y}) = \frac{1}{n_\text{samples}} \sum_{i=0}^{n_\text{samples}-1} 1(\hat{y}_i = y_i)$$

where $1(x)$ is the indicator function.

```
>>> import numpy as np
>>> from sklearn.metrics import accuracy_score
>>> y_pred = [0, 2, 1, 3]
>>> y_true = [0, 1, 2, 3]
>>> accuracy_score(y_true, y_pred)
0.5
>>> accuracy_score(y_true, y_pred, normalize=False)
2.0
```

In the multilabel case with binary label indicators:

```
>>> accuracy_score(np.array([[0, 1], [1, 1]]), np.ones((2, 2)))
0.5
```

Examples

- See Test with permutations of the significance of a classification score for an example of accuracy score usage using permutations of the dataset.

3.4.4.3. Top-k accuracy score

The **`top_k_accuracy_score`** function is a generalization of **`accuracy_score`**. The difference is that a prediction is considered correct as long as the true label is associated with one of the `k` highest predicted scores. **`accuracy_score`** is the special case of `k = 1`.

The function covers the binary and multiclass classification cases but not the multilabel case.

If $\hat{f}_{i,j}$ is the predicted class for the i-th sample corresponding to the j-th largest predicted score and y_i is the corresponding true value, then the fraction of correct predictions over n_samples is defined as

$$\texttt{top-k accuracy}(y, \hat{f}) = \frac{1}{n_\text{samples}} \sum_{i=0}^{n_\text{samples}-1} \sum_{j=1}^{k} 1(\hat{f}_{i,j} = y_i)$$

where k is the number of guesses allowed and $1(x)$ is the indicator function.

```
>>> import numpy as np
>>> from sklearn.metrics import top_k_accuracy_score
>>> y_true = np.array([0, 1, 2, 2])
>>> y_score = np.array([[0.5, 0.2, 0.2],
...                     [0.3, 0.4, 0.2],
...                     [0.2, 0.4, 0.3],
...                     [0.7, 0.2, 0.1]])
>>> top_k_accuracy_score(y_true, y_score, k=2)
0.75
>>> # Not normalizing gives the number of "correctly" classified samples
>>> top_k_accuracy_score(y_true, y_score, k=2, normalize=False)
3.0
```

3.4.4.4. Balanced accuracy score

The **`balanced_accuracy_score`** function computes the balanced accuracy, which avoids inflated performance estimates on imbalanced datasets. It is the macro-average of recall scores per class or, equivalently, raw accuracy where each sample is weighted according to the inverse prevalence of its true class. Thus for balanced datasets, the score is equal to accuracy.

In the binary case, balanced accuracy is equal to the arithmetic mean of sensitivity (true positive rate) and specificity (true negative rate), or the area under the ROC curve with binary predictions rather than scores:

$$\texttt{balanced-accuracy} = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right)$$

If the classifier performs equally well on either class, this term reduces to the conventional accuracy (i.e., the number of correct predictions divided by the total number of predictions).

In contrast, if the conventional accuracy is above chance only because the classifier takes advantage of an imbalanced test set, then the balanced accuracy, as appropriate, will drop to $\frac{1}{n\_classes}$.

The score ranges from 0 to 1, or when `adjusted=True` is used, it is rescaled to the range $\frac{1}{1-n\_classes}$ to 1, inclusive, with performance at random scoring 0.

If y_i is the true value of the i-th sample, and w_i is the corresponding sample weight, then we adjust the sample weight to:

$$\hat{w}_i = \frac{w_i}{\sum_j 1(y_j = y_i) w_j}$$

where $1(x)$ is the indicator function. Given predicted $\hat{y}_i$ for sample i, balanced accuracy is defined as:

$$\texttt{balanced-accuracy}(y, \hat{y}, w) = \frac{1}{\sum \hat{w}_i} \sum_i 1(\hat{y}_i = y_i) \hat{w}_i$$

With `adjusted=True`, balanced accuracy reports the relative increase from $\texttt{balanced-accuracy}(y, \mathbf{0}, w) = \frac{1}{n\_classes}$. In the binary case, this is also known as Youden's J statistic, or *informedness*.

Note

The multiclass definition here seems the most reasonable extension of the metric used in binary classification, though there is no certain consensus in the literature:

- Our definition: [Mosley2013], [Kelleher2015] and [Guyon2015], where [Guyon2015] adopt the adjusted version to ensure that random predictions have a score of 0 and perfect predictions have a score of 1.
- Class balanced accuracy as described in [Mosley2013]: the minimum between the precision and the recall for each class is computed. Those values are then averaged over the total number of classes to get the balanced accuracy.
- Balanced Accuracy as described in [Urbanowicz2015]: the average of sensitivity and specificity is computed for each class and then averaged over total number of classes.

References

[Guyon2015](1,2) I. Guyon, K. Bennett, G. Cawley, H.J. Escalante, S. Escalera, T.K. Ho, N. Macià, B. Ray, M. Saeed, A.R. Statnikov, E. Viegas, Design of the 2015 ChaLearn AutoML Challenge, IJCNN 2015.
[Mosley2013](1,2) L. Mosley, A balanced approach to the multi-class imbalance problem, IJCV 2010.
[Kelleher2015] John. D. Kelleher, Brian Mac Namee, Aoife D'Arcy, Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies, 2015.
[Urbanowicz2015] Urbanowicz R.J., Moore, J.H. ExSTraCS 2.0: description and evaluation of a scalable learning classifier system, Evol. Intel. (2015) 8: 89.

3.4.4.5. Cohen's kappa

The function **`cohen_kappa_score`** computes Cohen's kappa statistic. This measure is intended to compare labelings by different human annotators, not a classifier versus a ground truth.

The kappa score is a number between -1 and 1. Scores above .8 are generally considered good agreement; zero or lower means no agreement (practically random labels).

Kappa scores can be computed for binary or multiclass problems, but not for multilabel problems (except by manually computing a per-label score) and not for more than two annotators.

```
>>> from sklearn.metrics import cohen_kappa_score
>>> labeling1 = [2, 0, 2, 2, 0, 1]
>>> labeling2 = [0, 0, 2, 2, 0, 2]
>>> cohen_kappa_score(labeling1, labeling2)
0.4285714285714286
```

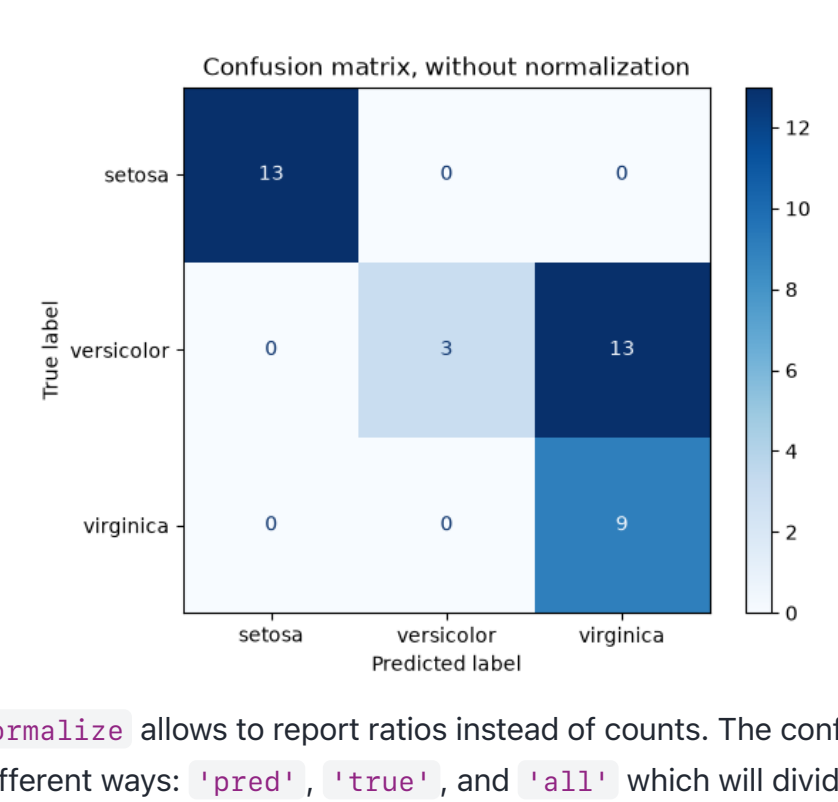
3.4.4.6. Confusion matrix

The **`confusion_matrix`** function evaluates classification accuracy by computing the confusion matrix with each row corresponding to the true class (Wikipedia and other references may use different convention for axes).

By definition, entry i, j in a confusion matrix is the number of observations actually in group i, but predicted to be in group j. Here is an example:

```
>>> from sklearn.metrics import confusion_matrix
>>> y_true = [2, 0, 2, 2, 0, 1]
>>> y_pred = [0, 0, 2, 2, 0, 2]
>>> confusion_matrix(y_true, y_pred)
array([[2, 0, 0],
       [0, 0, 1],
       [1, 0, 2]])
```

`ConfusionMatrixDisplay` can be used to visually represent a confusion matrix as shown in the Evaluate the performance of a classifier with Confusion Matrix example, which creates the following figure:



The parameter `normalize` allows to report ratios instead of counts. The confusion matrix can be normalized in 3 different ways: `'pred'`, `'true'`, and `'all'` which will divide the counts by the sum of each columns, rows, or the entire matrix, respectively.

```
>>> y_true = [0, 0, 0, 1, 1, 1, 1, 1]
>>> y_pred = [0, 1, 0, 1, 0, 1, 0, 1]
>>> confusion_matrix(y_true, y_pred, normalize='all')
array([[0.25 , 0.125],
       [0.25 , 0.375]])
```

For binary problems, we can get counts of true negatives, false positives, false negatives and true positives as follows:

```
>>> y_true = [0, 0, 0, 1, 1, 1, 1, 1]
>>> y_pred = [0, 1, 0, 1, 0, 1, 0, 1]
>>> tn, fp, fn, tp = confusion_matrix(y_true, y_pred).ravel().tolist()
>>> tn, fp, fn, tp
(2, 1, 2, 3)
```

With **`confusion_matrix_at_thresholds`** we can get true negatives, false positives, false negatives and true positives for different thresholds:

```
>>> from sklearn.metrics import confusion_matrix_at_thresholds
>>> y_true = np.array([0., 0., 1., 1.])
>>> y_score = np.array([0.1, 0.4, 0.35, 0.8])
>>> tns, fps, fns, tps, thresholds = confusion_matrix_at_thresholds(y_true, y_score)
>>> tns
array([2., 1., 1., 0.])
>>> fps
array([0., 1., 1., 2.])
>>> fns
array([1., 1., 0., 0.])
>>> tps
array([1., 1., 2., 2.])
>>> thresholds
array([0.8, 0.4, 0.35, 0.1])
```

Note that the thresholds consist of distinct `y_score` values, in decreasing order.

Examples

- See Evaluate the performance of a classifier with Confusion Matrix for an example of using a confusion matrix to evaluate classifier output quality.
- See Recognizing hand-written digits for an example of using a confusion matrix to classify hand-written digits.
- See Classification of text documents using sparse features for an example of using a confusion matrix to classify text documents.

3.4.4.7. Classification report

The **`classification_report`** function builds a text report showing the main classification metrics. Here is a small example with custom `target_names` and inferred labels:

```
>>> from sklearn.metrics import classification_report
>>> y_true = [0, 1, 2, 2, 0]
>>> y_pred = [0, 0, 2, 1, 0]
>>> target_names = ['class 0', 'class 1', 'class 2']
>>> print(classification_report(y_true, y_pred, target_names=target_names))
              precision    recall  f1-score   support

     class 0       0.67      1.00      0.80         2
     class 1       0.00      0.00      0.00         1
     class 2       1.00      0.50      0.67         2

    accuracy                           0.60         5
   macro avg       0.56      0.50      0.49         5
weighted avg       0.67      0.60      0.59         5
```

Examples

- See Recognizing hand-written digits for an example of classification report usage for hand-written digits.
- See Custom refit strategy of a grid search with cross-validation for an example of classification report usage for grid search with nested cross-validation.

3.4.4.8. Hamming loss

The **`hamming_loss`** computes the average Hamming loss or Hamming distance between two sets of samples.

If $\hat{y}_{i,j}$ is the predicted value for the j-th label of a given sample i, $y_{i,j}$ is the corresponding true value, n_samples is the number of samples and n_labels is the number of labels, then the Hamming loss $L_{Hamming}$ is defined as:

$$L_{Hamming}(y, \hat{y}) = \frac{1}{n_\text{samples} * n_\text{labels}} \sum_{i=0}^{n_\text{samples}-1} \sum_{j=0}^{n_\text{labels}-1} 1(\hat{y}_{i,j} \not= y_{i,j})$$

where $1(x)$ is the indicator function.

The equation above does not hold true in the case of multiclass classification. Please refer to the note below for more information.

```
>>> from sklearn.metrics import hamming_loss
>>> y_pred = [1, 2, 3, 4]
>>> y_true = [2, 2, 3, 4]
>>> hamming_loss(y_true, y_pred)
0.25
```

In the multilabel case with binary label indicators:

```
>>> hamming_loss(np.array([[0, 1], [1, 1]]), np.zeros((2, 2)))
0.75
```

Note

In multiclass classification, the Hamming loss corresponds to the Hamming distance between `y_true` and `y_pred` which is similar to the Zero one loss function. However, while zero-one loss penalizes prediction sets that do not strictly match true sets, the Hamming loss penalizes individual labels. Thus the Hamming loss, upper bounded by the zero-one loss, is always between zero and one, inclusive; and predicting a proper subset or superset of the true labels will give a Hamming loss between zero and one, exclusive.

3.4.4.9. Precision, recall and F-measures

Intuitively, precision is the ability of the classifier not to label as positive a sample that is negative, and recall is the ability of the classifier to find all the positive samples.

The F-measure (F_β and F_1 measures) can be interpreted as a weighted harmonic mean of the precision and recall. A F_β measure reaches its best value at 1 and its worst score at 0. With $\beta = 1$, F_β and F_1 are equivalent, and the recall and the precision are equally important.

The **`precision_recall_curve`** computes a precision-recall curve from the ground truth label and a score given by the classifier by varying a decision threshold.

The **`average_precision_score`** function computes the average precision (AP) from prediction scores. The value is between 0 and 1 and higher is better. AP is defined as

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n$$

where P_n and R_n are the precision and recall at the nth threshold. With random predictions, the AP is the fraction of positive samples.

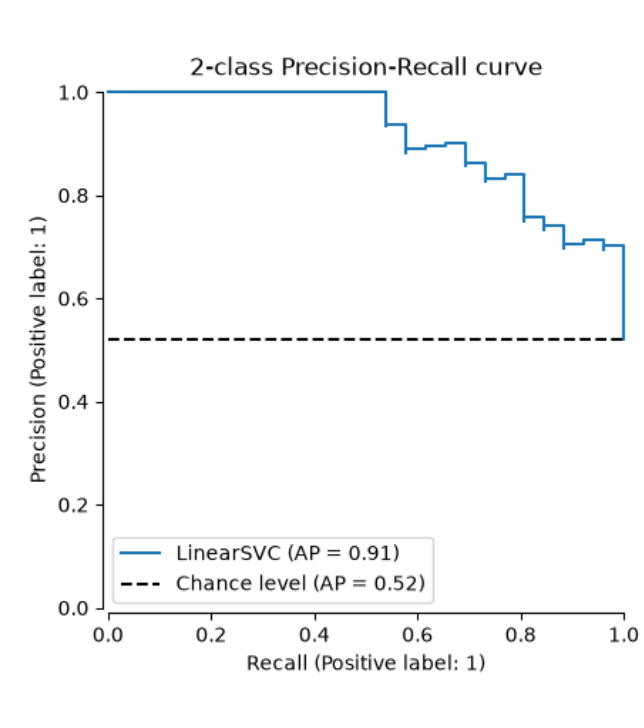
References [Manning2008] and [Everingham2010] present alternative variants of AP that interpolate the precision-recall curve. Currently, **`average_precision_score`** does not implement any interpolated variant. References [Davis2006] and [Flach2015] describe why a linear interpolation of points on the precision-recall curve provides an overly-optimistic measure of classifier performance. This linear interpolation is used when computing area under the curve with the trapezoidal rule in **`auc`**. [Chen2024] benchmarks different interpolation strategies to demonstrate the effects.

Several functions allow you to analyze the precision, recall and F-measures score:

`average_precision_score` (y_true, y_score, *)	Compute average precision (AP) from prediction scores.
`f1_score` (y_true, y_pred, *[, labels, ...])	Compute the F1 score, also known as balanced F-score or F-measure.
`fbeta_score` (y_true, y_pred, *, beta[, ...])	Compute the F-beta score.
`precision_recall_curve` (y_true, y_score, *[, ...])	Compute precision-recall pairs for different probability thresholds.
`precision_recall_fscore_support` (y_true, ...)	Compute precision, recall, F-measure and support for each class.
`precision_score` (y_true, y_pred, *[, labels, ...])	Compute the precision.
`recall_score` (y_true, y_pred, *[, labels, ...])	Compute the recall.

Note that the **`precision_recall_curve`** function is restricted to the binary case. The **`average_precision_score`** function supports multiclass and multilabel formats by computing each class score in a One-vs-the-rest (OvR) fashion and averaging them or not depending of its `average` argument value.

The **`PrecisionRecallDisplay.from_estimator`** and **`PrecisionRecallDisplay.from_predictions`** functions will plot the precision-recall curve as follows.



Examples

- See Custom refit strategy of a grid search with cross-validation for an example of **`precision_score`** and **`recall_score`** usage to estimate parameters using grid search with nested cross-validation.
- See Precision-Recall for an example of **`precision_recall_curve`** usage to evaluate classifier output quality.

References

[Manning2008] C.D. Manning, P. Raghavan, H. Schütze, Introduction to Information Retrieval, 2008.
[Everingham2010] M. Everingham, L. Van Gool, C.K.I. Williams, J. Winn, A. Zisserman, The Pascal Visual Object Classes (VOC) Challenge, IJCV 2010.
[Davis2006] J. Davis, M. Goadrich, The Relationship Between Precision-Recall and ROC Curves, ICML 2006.
[Flach2015] P.A. Flach, M. Kull, Precision-Recall-Gain Curves: PR Analysis Done Right, NIPS 2015.
[Chen2024] W. Chen, C. Miao, Z. Zhang, C.S. Fung, R. Wang, Y. Chen, Y. Qian, L. Cheng, K.Y. Yip, S.K Tsui, Q. Cao, Commonly used software tools produce conflicting and overly-optimistic AUPRC values, Genome Biology 2024.

3.4.4.9.1. Binary classification

In a binary classification task, the terms ''positive'' and ''negative'' refer to the classifier's prediction, and the terms ''true'' and ''false'' refer to whether that prediction corresponds to the external judgment (sometimes known as the ''observation''). Given these definitions, we can formulate the following table:

	Actual class (observation)	
Predicted class (expectation)	tp (true positive) Correct result	fp (false positive) Unexpected result
	fn (false negative) Missing result	tn (true negative) Correct absence of result

In this context, we can define the notions of precision and recall:

$$\text{precision} = \frac{\text{tp}}{\text{tp} + \text{fp}},$$

$$\text{recall} = \frac{\text{tp}}{\text{tp} + \text{fn}},$$

(Sometimes recall is also called ''sensitivity'')

F-measure is the weighted harmonic mean of precision and recall, with precision's contribution to the mean weighted by some parameter β:

$$F_\beta = (1 + \beta^2) \frac{\text{precision} \times \text{recall}}{\beta^2 \text{precision} + \text{recall}}$$

To avoid division by zero when precision and recall are zero, Scikit-Learn calculates F-measure with this otherwise-equivalent formula:

$$F_\beta = \frac{(1 + \beta^2) \text{tp}}{(1 + \beta^2) \text{tp} + \text{fp} + \beta^2 \text{fn}}$$

Note that this formula is still undefined when there are no true positives, false positives, or false negatives. By default, F-1 for a set of exclusively true negatives is calculated as 0, however this behavior can be changed using the `zero_division` parameter. Here are some small examples in binary classification:

```
>>> from sklearn import metrics
>>> y_pred = [0, 1, 0, 0]
>>> y_true = [0, 1, 0, 1]
>>> metrics.precision_score(y_true, y_pred)
1.0
>>> metrics.recall_score(y_true, y_pred)
0.5
>>> metrics.f1_score(y_true, y_pred)
0.66
>>> metrics.fbeta_score(y_true, y_pred, beta=0.5)
0.83
>>> metrics.fbeta_score(y_true, y_pred, beta=1)
0.66
>>> metrics.fbeta_score(y_true, y_pred, beta=2)
0.55
>>> metrics.precision_recall_fscore_support(y_true, y_pred, beta=0.5)
(array([0.66, 1.        ]), array([1. , 0.5]), array([0.71, 0.83]), array([2, 2]))


>>> import numpy as np
>>> from sklearn.metrics import precision_recall_curve
>>> from sklearn.metrics import average_precision_score
>>> y_true = np.array([0, 0, 1, 1])
>>> y_scores = np.array([0.1, 0.4, 0.35, 0.8])
>>> precision, recall, threshold = precision_recall_curve(y_true, y_scores)
>>> precision
array([0.5       , 0.66, 0.5       , 1.        , 1.        ])
>>> recall
array([1. , 1. , 0.5, 0.5, 0. ])
>>> threshold
array([0.1 , 0.35, 0.4 , 0.8 ])
>>> average_precision_score(y_true, y_scores)
0.83
```

3.4.4.9.2. Multiclass and multilabel classification

In a multiclass and multilabel classification task, the notions of precision, recall, and F-measures can be applied to each label independently. There are a few ways to combine results across labels, specified by the `average` argument to the **`average_precision_score`**, **`f1_score`**, **`fbeta_score`**, **`precision_recall_fscore_support`**, **`precision_score`** and **`recall_score`** functions, as described above.

Note the following behaviors when averaging:

- If all labels are included, "micro"-averaging in a multiclass setting will produce precision, recall and F that are all identical to accuracy.
- "weighted" averaging may produce an F-score that is not between precision and recall.
- "macro" averaging for F-measures is calculated as the arithmetic mean over per-label/class F-measures, not the harmonic mean over the arithmetic precision and recall means. Both calculations can be seen in the literature but are not equivalent, see [OB2019] for details.

To make this more explicit, consider the following notation:

- y the set of *true* $(sample, label)$ pairs
- $\hat{y}$ the set of *predicted* $(sample, label)$ pairs
- L the set of labels
- S the set of samples
- y_s the subset of y with sample s, i.e. $y_s := \left\{(s', l) \in y | s' = s\right\}$
- y_l the subset of y with label l
- similarly, $\hat{y}_s$ and $\hat{y}_l$ are subsets of $\hat{y}$
- $P(A, B) := \frac{\left| A \cap B \right|}{\left|B\right|}$ for some sets A and B
- $R(A, B) := \frac{\left| A \cap B \right|}{\left|A\right|}$ (Conventions vary on handling $A = \emptyset$; this implementation uses $R(A, B):=0$, and similar for P.)
- $F_\beta(A, B) := \left(1 + \beta^2\right) \frac{P(A, B) \times R(A, B)}{\beta^2 P(A, B) + R(A, B)}$

Then the metrics are defined as:

`average`	Precision	Recall	F_beta
`"micro"`	$P(y, \hat{y})$	$R(y, \hat{y})$	$F_\beta(y, \hat{y})$
`"samples"`	$\frac{1}{\left|S\right|} \sum_{s \in S} P(y_s, \hat{y}_s)$	$\frac{1}{\left|S\right|} \sum_{s \in S} R(y_s, \hat{y}_s)$	$\frac{1}{\left|S\right|} \sum_{s \in S} F_\beta(y_s, \hat{y}_s)$
`"macro"`	$\frac{1}{\left|L\right|} \sum_{l \in L} P(y_l, \hat{y}_l)$	$\frac{1}{\left|L\right|} \sum_{l \in L} R(y_l, \hat{y}_l)$	$\frac{1}{\left|L\right|} \sum_{l \in L} F_\beta(y_l, \hat{y}_l)$
`"weighted"`	$\frac{1}{\sum_{l \in L} \left|y_l\right|} \sum_{l \in L} \left|y_l\right| P(y_l, \hat{y}_l)$	$\frac{1}{\sum_{l \in L} \left|y_l\right|} \sum_{l \in L} \left|y_l\right| R(y_l, \hat{y}_l)$	$\frac{1}{\sum_{l \in L} \left|y_l\right|} \sum_{l \in L} \left|y_l\right| F_\beta(y_l, \hat{y}_l)$
`None`	$\langle P(y_l, \hat{y}_l) | l \in L \rangle$	$\langle R(y_l, \hat{y}_l) | l \in L \rangle$	$\langle F_\beta(y_l, \hat{y}_l) | l \in L \rangle$

```
>>> from sklearn import metrics
>>> y_true = [0, 1, 2, 0, 1, 2]
>>> y_pred = [0, 2, 1, 0, 0, 1]
>>> metrics.precision_score(y_true, y_pred, average='macro')
0.22
>>> metrics.recall_score(y_true, y_pred, average='micro')
0.33
>>> metrics.f1_score(y_true, y_pred, average='weighted')
0.267
>>> metrics.fbeta_score(y_true, y_pred, average='macro', beta=0.5)
0.238
>>> metrics.precision_recall_fscore_support(y_true, y_pred, beta=0.5, average=None)
(array([0.667, 0., 0.]), array([1., 0., 0.]), array([0.714, 0., 0.]), array([2, 2, 2]
```

For multiclass classification with a "negative class", it is possible to exclude some labels:

```
>>> metrics.recall_score(y_true, y_pred, labels=[1, 2], average='micro')
... # excluding 0, no labels were correctly recalled
0.0
```

Similarly, labels not present in the data sample may be accounted for in macro-averaging.

```
>>> metrics.precision_score(y_true, y_pred, labels=[0, 1, 2, 3], average='macro')
0.166
```

References

[OB2019] Opitz, J., & Burst, S. (2019). "Macro f1 and macro f1."

3.4.4.10. Jaccard similarity coefficient score

The **`jaccard_score`** function computes the average of Jaccard similarity coefficients, also called the Jaccard index, between pairs of label sets.

The Jaccard similarity coefficient with a ground truth label set y and predicted label set $\hat{y}$, is defined as

$$J(y, \hat{y}) = \frac{|y \cap \hat{y}|}{|y \cup \hat{y}|}.$$

The **`jaccard_score`** (like **`precision_recall_fscore_support`**) applies natively to binary targets. By computing it set-wise it can be extended to apply to multilabel and multiclass through the use of `average` (see above).

In the binary case:

```
>>> import numpy as np
>>> from sklearn.metrics import jaccard_score
>>> y_true = np.array([[0, 1, 1],
...                    [1, 1, 0]])
>>> y_pred = np.array([[1, 1, 1],
...                    [1, 0, 0]])
>>> jaccard_score(y_true[0], y_pred[0])
0.6666
```

In the 2D comparison case (e.g. image similarity):

```
>>> jaccard_score(y_true, y_pred, average="micro")
0.6
```

In the multilabel case with binary label indicators:

```
>>> jaccard_score(y_true, y_pred, average='samples')
0.5833
>>> jaccard_score(y_true, y_pred, average='macro')
0.6666
>>> jaccard_score(y_true, y_pred, average=None)
array([0.5, 0.5, 1. ])
```

Multiclass problems are binarized and treated like the corresponding multilabel problem:

```
>>> y_pred = [0, 2, 1, 2]
>>> y_true = [0, 1, 2, 2]
>>> jaccard_score(y_true, y_pred, average=None)
array([1. , 0. , 0.33])
>>> jaccard_score(y_true, y_pred, average='macro')
0.44
>>> jaccard_score(y_true, y_pred, average='micro')
0.33
```

3.4.4.11. Hinge loss

The **`hinge_loss`** function computes the average distance between the model and the data using hinge loss, a one-sided metric that considers only prediction errors. (Hinge loss is used in maximal margin classifiers such as support vector machines.)

If the true label y_i of a binary classification task is encoded as $y_i=\left\{-1, +1\right\}$ for every sample i; and w_i is the corresponding predicted decision (an array of shape (`n_samples`,) as output by the `decision_function` method), then the hinge loss is defined as:

$$L_\text{Hinge}(y, w) = \frac{1}{n_\text{samples}} \sum_{i=0}^{n_\text{samples}-1} \max\left\{1 - w_i y_i, 0\right\}$$

If there are more than two labels, **`hinge_loss`** uses a multiclass variant due to Crammer & Singer. Here is the paper describing it.

In this case the predicted decision is an array of shape (`n_samples`, `n_labels`). If w_{i, y_i} is the predicted decision for the true label y_i of the i-th sample; and $\hat{w}_{i, y_i} = \max\left\{w_{i, y_j}~|~y_j \ne y_i \right\}$ is the maximum of the predicted decisions for all the other labels, then the multi-class hinge loss is defined by:

$$L_\text{Hinge}(y, w) = \frac{1}{n_\text{samples}} \sum_{i=0}^{n_\text{samples}-1} \max\left\{1 + \hat{w}_{i, y_i} - w_{i, y_i}, 0\right\}$$

Here is a small example demonstrating the use of the **`hinge_loss`** function with an svm classifier in a binary class problem:

```
>>> from sklearn import svm
>>> from sklearn.metrics import hinge_loss
>>> X = [[0], [1]]
>>> y = [-1, 1]
>>> est = svm.LinearSVC(random_state=0)
>>> est.fit(X, y)
LinearSVC(random_state=0)
>>> pred_decision = est.decision_function([[-2], [3], [0.5]])
>>> pred_decision
array([-2.18,  2.36,  0.09])
>>> hinge_loss([-1, 1, 1], pred_decision)
0.3
```

Here is an example demonstrating the use of the **`hinge_loss`** function with an svm classifier in a multiclass problem:

```
>>> X = np.array([[0], [1], [2], [3]])
>>> Y = np.array([0, 1, 2, 3])
>>> labels = np.array([0, 1, 2, 3])
>>> est = svm.LinearSVC()
>>> est.fit(X, Y)
LinearSVC()
>>> pred_decision = est.decision_function([[-1], [2], [3]])
>>> y_true = [0, 2, 3]
>>> hinge_loss(y_true, pred_decision, labels=labels)
0.56
```

3.4.4.12. Log loss

Log loss, also called logistic regression loss or cross-entropy loss, is defined on probability estimates. It is commonly used in (multinomial) logistic regression and neural networks, as well as in some variants of expectation-maximization, and can be used to evaluate the probability outputs (`predict_proba`) of a classifier instead of its discrete predictions.

For binary classification with a true label $y \in \{0,1\}$ and a probability estimate $\hat{p} \approx \operatorname{Pr}(y = 1)$, the log loss per sample is the negative log-likelihood of the classifier given the true label:

$$L_{\log}(y, \hat{p}) = -\log \operatorname{Pr}(y|\hat{p}) = -(y \log (\hat{p}) + (1 - y) \log (1 - \hat{p}))$$

This extends to the multiclass case as follows. Let the true labels for a set of samples be encoded as a 1-of-K binary indicator matrix Y, i.e., $y_{i,k} = 1$ if sample i has label k taken from a set of K labels. Let $\hat{P}$ be a matrix of probability estimates, with elements $\hat{p}_{i,k} \approx \operatorname{Pr}(y_{i,k} = 1)$. Then the log loss of the whole set is

$$L_{\log}(Y, \hat{P}) = -\log \operatorname{Pr}(Y|\hat{P}) = - \frac{1}{N} \sum_{i=0}^{N-1} \sum_{k=0}^{K-1} y_{i,k} \log \hat{p}_{i,k}$$

To see how this generalizes the binary log loss given above, note that in the binary case, $\hat{p}_{i,0} = 1 - \hat{p}_{i,1}$ and $y_{i,0} = 1 - y_{i,1}$, so expanding the inner sum over $y_{i,k} \in \{0,1\}$ gives the binary log loss.

The **`log_loss`** function computes log loss given a list of ground-truth labels and a probability matrix, as returned by an estimator's `predict_proba` method.

```
>>> from sklearn.metrics import log_loss
>>> y_true = [0, 0, 1, 1]
>>> y_proba = [[.9, .1], [.8, .2], [.3, .7], [.01, .99]]
>>> log_loss(y_true, y_proba)
0.1738
```

The first `[.9, .1]` in `y_proba` denotes 90% probability that the first sample has label 0. The log loss is non-negative.

3.4.4.13. Matthews correlation coefficient

The **`matthews_corrcoef`** function computes the Matthew's correlation coefficient (MCC) for binary classes. Quoting Wikipedia:

> "The Matthews correlation coefficient is used in machine learning as a measure of the quality of binary (two-class) classifications. It takes into account true and false positives and negatives and is generally regarded as a balanced measure which can be used even if the classes are of very different sizes. The MCC is in essence a correlation coefficient value between -1 and +1. A coefficient of +1 represents a perfect prediction, 0 an average random prediction and -1 an inverse prediction. The statistic is also known as the phi coefficient."

In the binary (two-class) case, tp, tn, fp and fn are respectively the number of true positives, true negatives, false positives and false negatives, the MCC is defined as

$$MCC = \frac{tp \times tn - fp \times fn}{\sqrt{(tp + fp)(tp + fn)(tn + fp)(tn + fn)}}.$$

In the multiclass case, the Matthews correlation coefficient can be defined in terms of a **`confusion_matrix`** C for K classes. To simplify the definition consider the following intermediate variables:

- $t_k=\sum_{i}^{K} C_{ik}$ the number of times class k truly occurred,
- $p_k=\sum_{i}^{K} C_{ki}$ the number of times class k was predicted,
- $c=\sum_{k}^{K} C_{kk}$ the total number of samples correctly predicted,
- $s=\sum_{i}^{K} \sum_{j}^{K} C_{ij}$ the total number of samples.

Then the multiclass MCC is defined as:

$$MCC = \frac{ c \times s - \sum_{k}^{K} p_k \times t_k }{\sqrt{ (s^2 - \sum_{k}^{K} p_k^2) \times (s^2 - \sum_{k}^{K} t_k^2) }}$$

When there are more than two labels, the value of the MCC will no longer range between -1 and +1. Instead the minimum value will be somewhere between -1 and 0 depending on the number and distribution of ground truth labels. The maximum value is always +1. For additional information, see [WikipediaMCC2021].

Here is a small example illustrating the usage of the **`matthews_corrcoef`** function:

```
>>> from sklearn.metrics import matthews_corrcoef
>>> y_true = [+1, +1, +1, -1]
>>> y_pred = [+1, -1, +1, +1]
>>> matthews_corrcoef(y_true, y_pred)
-0.33
```

References

[WikipediaMCC2021] Wikipedia contributors. Phi coefficient. Wikipedia, The Free Encyclopedia. April 21, 2021, 12:21 CEST. Available at: https://en.wikipedia.org/wiki/Phi_coefficient Accessed April 21, 2021.

3.4.4.14. Multi-label confusion matrix

The **`multilabel_confusion_matrix`** function computes class-wise (default) or sample-wise (samplewise=True) multilabel confusion matrix to evaluate the accuracy of a classification. multilabel_confusion_matrix also treats multiclass data as if it were multilabel, as this is a transformation commonly applied to evaluate multiclass problems with binary classification metrics (such as precision, recall, etc.).

When calculating class-wise multilabel confusion matrix C, the count of true negatives for class i is $C_{i,0,0}$, false negatives is $C_{i,1,0}$, true positives is $C_{i,1,1}$ and false positives is $C_{i,0,1}$.

Here is an example demonstrating the use of the **`multilabel_confusion_matrix`** function with multilabel indicator matrix input:

```
>>> import numpy as np
>>> from sklearn.metrics import multilabel_confusion_matrix
>>> y_true = np.array([[1, 0, 1],
...                    [0, 1, 0]])
>>> y_pred = np.array([[1, 0, 0],
...                    [0, 1, 1]])
>>> multilabel_confusion_matrix(y_true, y_pred)
array([[[1, 0],
        [0, 1]],

       [[1, 0],
        [0, 1]],

       [[0, 1],
        [1, 0]]])
```

Or a confusion matrix can be constructed for each sample's labels:

```
>>> multilabel_confusion_matrix(y_true, y_pred, samplewise=True)
array([[[1, 0],
        [1, 1]],

       [[1, 1],
        [0, 1]]])
```

Here is an example demonstrating the use of the **`multilabel_confusion_matrix`** function with multiclass input:

```
>>> y_true = ["cat", "ant", "cat", "cat", "ant", "bird"]
>>> y_pred = ["ant", "ant", "cat", "cat", "ant", "cat"]
>>> multilabel_confusion_matrix(y_true, y_pred,
...                             labels=["ant", "bird", "cat"])
array([[[3, 1],
        [0, 2]],

       [[5, 0],
        [1, 0]],

       [[2, 1],
        [1, 2]]])
```

Here are some examples demonstrating the use of the **`multilabel_confusion_matrix`** function to calculate recall (or sensitivity), specificity, fall out and miss rate for each class in a problem with multilabel indicator matrix input.

Calculating recall (also called the true positive rate or the sensitivity) for each class:

```
>>> y_true = np.array([[0, 0, 1],
...                    [0, 1, 0],
...                    [1, 1, 0]])
>>> y_pred = np.array([[0, 1, 0],
...                    [0, 0, 1],
...                    [1, 1, 0]])
>>> mcm = multilabel_confusion_matrix(y_true, y_pred)
>>> tn = mcm[:, 0, 0]
>>> tp = mcm[:, 1, 1]
>>> fn = mcm[:, 1, 0]
>>> fp = mcm[:, 0, 1]
>>> tp / (tp + fn)
array([1. , 0.5, 0. ])
```

Calculating specificity (also called the true negative rate) for each class:

```
>>> tn / (tn + fp)
array([1. , 0. , 0.5])
```

Calculating fall out (also called the false positive rate) for each class:

```
>>> fp / (fp + tn)
array([0. , 1. , 0.5])
```

Calculating miss rate (also called the false negative rate) for each class:

```
>>> fn / (fn + tp)
array([0. , 0.5, 1. ])
```

3.4.4.15. Receiver operating characteristic (ROC)

The function **`roc_curve`** computes the receiver operating characteristic curve, or ROC curve. Quoting Wikipedia :

> "A receiver operating characteristic (ROC), or simply ROC curve, is a graphical plot which illustrates the performance of a binary classifier system as its discrimination threshold is varied. It is created by plotting the fraction of true positives out of the positives (TPR = true positive rate) vs. the fraction of false positives out of the negatives (FPR = false positive rate), at various threshold settings. TPR is also known as sensitivity, and FPR is one minus the specificity or true negative rate."

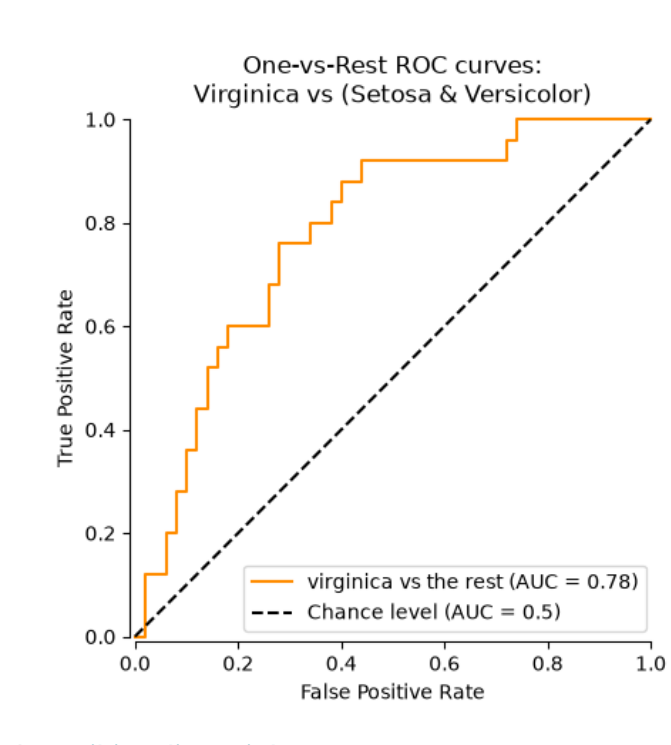
This function requires the true binary value and the target scores, which can either be probability estimates of the positive class or non-thresholded decision values (as returned by decision_function on some classifiers). Here is a small example of how to use the **`roc_curve`** function:

```
>>> import numpy as np
>>> from sklearn.metrics import roc_curve
>>> y = np.array([1, 1, 2, 2])
>>> scores = np.array([0.1, 0.4, 0.35, 0.8])
>>> fpr, tpr, thresholds = roc_curve(y, scores, pos_label=2)
>>> fpr
array([0. , 0. , 0.5, 0.5, 1. ])
>>> tpr
array([0. , 0.5, 0.5, 1. , 1. ])
>>> thresholds
array([ inf, 0.8 , 0.4 , 0.35, 0.1 ])
```

Compared to metrics such as the subset accuracy, the Hamming loss, or the F1 score, ROC doesn't require optimizing a threshold for each label.

The **`roc_auc_score`** function, denoted by ROC-AUC or AUROC, computes the area under the ROC curve. By doing so, the curve information is summarized in one number.

The following figure shows the ROC curve and ROC-AUC score for a classifier aimed to distinguish the virginica flower from the rest of the species in the Iris plants dataset:



For more information see the Wikipedia article on AUC.

3.4.4.15.1. Binary case

In the **binary case**, you can either provide the probability estimates, using the `classifier.predict_proba()` method, or the non-thresholded decision values given by the `classifier.decision_function()` method. In the case of providing the probability estimates, the probability of the class with the "greater label" should be provided. The "greater label" corresponds to `classifier.classes_[1]` and thus `classifier.predict_proba(X)[:, 1]`. Therefore, the `y_score` parameter is of size (n_samples,).

```
>>> from sklearn.datasets import load_breast_cancer
>>> from sklearn.linear_model import LogisticRegression
>>> from sklearn.metrics import roc_auc_score
>>> X, y = load_breast_cancer(return_X_y=True)
>>> clf = LogisticRegression().fit(X, y)
>>> clf.classes_
array([0, 1])
```

We can use the probability estimates corresponding to `clf.classes_[1]`.

```
>>> y_score = clf.predict_proba(X)[:, 1]
>>> roc_auc_score(y, y_score)
0.99
```

Otherwise, we can use the non-thresholded decision values

```
>>> roc_auc_score(y, clf.decision_function(X))
0.99
```

3.4.4.15.2. Multi-class case

The **`roc_auc_score`** function can also be used in **multi-class classification**. Two averaging strategies are currently supported: the one-vs-one algorithm computes the average of the pairwise ROC AUC scores, and the one-vs-rest algorithm computes the average of the ROC AUC scores for each class against all other classes. In both cases, the predicted labels are provided in an array with values from 0 to `n_classes`, and the scores correspond to the probability estimates that a sample belongs to a particular class. The OvO and OvR algorithms support weighting uniformly (`average='macro'`) and by prevalence (`average='weighted'`).

One-vs-one Algorithm

One-vs-rest Algorithm

3.4.4.15.3. Multi-label case

In **multi-label classification**, the **`roc_auc_score`** function is extended by averaging over the labels as above. In this case, you should provide a `y_score` of shape `(n_samples, n_classes)`. Thus, when using the probability estimates, one needs to select the probability of the class with the greater label for each output.

```
>>> from sklearn.datasets import make_multilabel_classification
>>> from sklearn.multioutput import MultiOutputClassifier
>>> X, y = make_multilabel_classification(random_state=0)
>>> inner_clf = LogisticRegression(random_state=0)
>>> clf = MultiOutputClassifier(inner_clf).fit(X, y)
>>> y_score = np.transpose([y_pred[:, 1] for y_pred in clf.predict_proba(X)])
>>> roc_auc_score(y, y_score, average=None)
array([0.828, 0.851, 0.94, 0.87, 0.95])
```

And the decision values do not require such processing.

```
>>> from sklearn.linear_model import RidgeClassifierCV
>>> clf = RidgeClassifierCV().fit(X, y)
>>> y_score = clf.decision_function(X)
>>> roc_auc_score(y, y_score, average=None)
array([0.82, 0.85, 0.93, 0.87, 0.94])
```

Examples

- See Multiclass Receiver Operating Characteristic (ROC) for an example of using ROC to evaluate the quality of the output of a classifier.
- See Receiver Operating Characteristic (ROC) with cross validation for an example of using ROC to evaluate classifier output quality, using cross-validation.
- See Species distribution modeling for an example of using ROC to model species distribution.

References

[HT2001](1,2) Hand, D.J. and Till, R.J., (2001). A simple generalisation of the area under the ROC curve for multiple class classification problems. Machine learning, 45(2), pp. 171-186.

[FC2009] Ferri, Cèsar & Hernandez-Orallo, Jose & Modroiu, R. (2009). An Experimental Comparison of Performance Measures for Classification. Pattern Recognition Letters. 30. 27-38.

[PD2000] Provost, F., Domingos, P. (2000). Well-trained PETs: Improving probability estimation trees (Section 6.2), CeDER Working Paper #IS-00-04, Stern School of Business, New York University.

[F2006] Fawcett, T., 2006. An introduction to ROC analysis. Pattern Recognition Letters, 27(8), pp. 861-874.

[F2001] Fawcett, T., 2001. Using rule sets to maximize ROC performance In Data Mining, 2001. Proceedings IEEE International Conference, pp. 131-138.

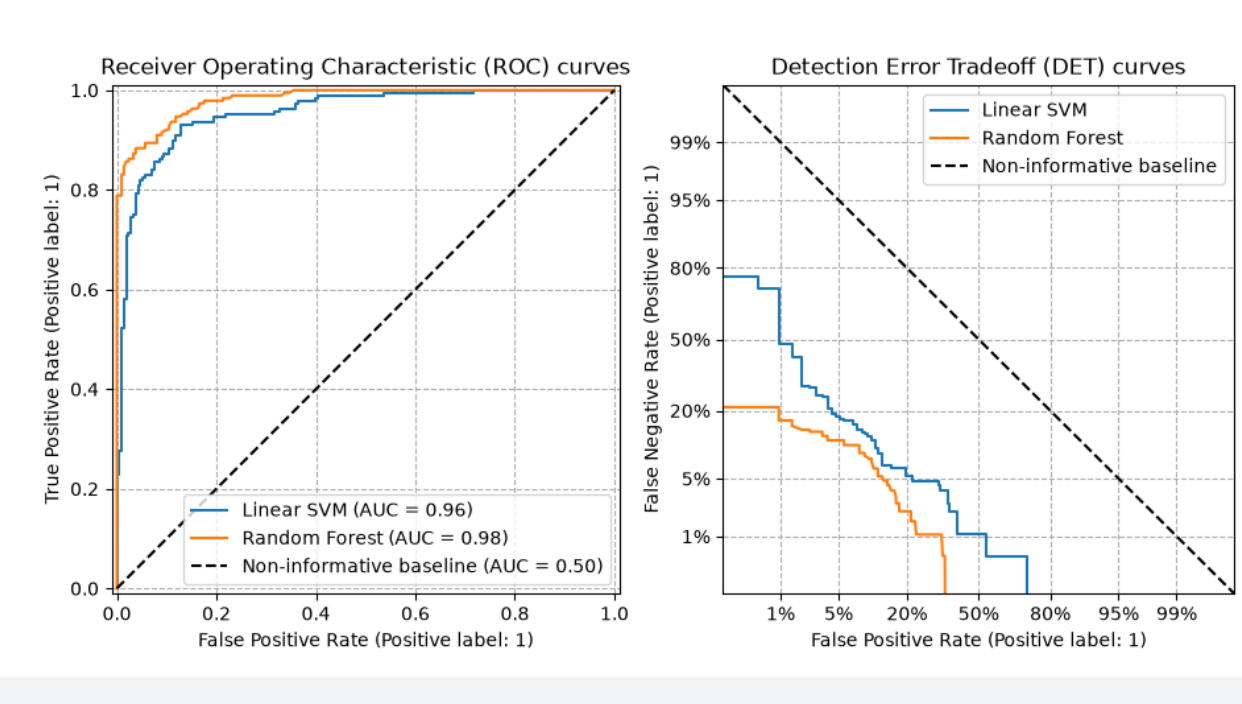
3.4.4.16. Detection error tradeoff (DET)

The function **`det_curve`** computes the detection error tradeoff curve (DET) curve [WikipediaDET2017]. Quoting Wikipedia:

> "A detection error tradeoff (DET) graph is a graphical plot of error rates for binary classification systems, plotting false reject rate vs. false accept rate. The x- and y-axes are scaled non-linearly by their standard normal deviates (or just by logarithmic transformation), yielding tradeoff curves that are more linear than ROC curves, and use most of the image area to highlight the differences of importance in the critical operating region."

DET curves are a variation of receiver operating characteristic (ROC) curves where False Negative Rate is plotted on the y-axis instead of True Positive Rate. DET curves are commonly plotted in normal deviate scale by transformation with ϕ^{-1} (with ϕ being the cumulative distribution function). The resulting performance curves explicitly visualize the tradeoff of error types for given classification algorithms. See [Martin1997] for examples and further motivation.

This figure compares the ROC and DET curves of two example classifiers on the same classification task:



Properties

Applications and limitations

Examples

- See Detection error tradeoff (DET) curve for an example comparison between receiver operating characteristic (ROC) curves and Detection error tradeoff (DET) curves.

References

[WikipediaDET2017] Wikipedia contributors. Detection error tradeoff. Wikipedia, The Free Encyclopedia. September 4, 2017, 23:33 UTC. Available at: https://en.wikipedia.org/w/index.php?title=Detection_error_tradeoff&oldid=798982054. Accessed February 19, 2018.

[Martin1997] A. Martin, G. Doddington, T. Kamm, M. Ordowski, and M. Przybocki, The DET Curve in Assessment of Detection Task Performance, NIST 1997.

[Navratil2007] J. Navratil and D. Klusacek, "On Linear DETs", 2007 IEEE International Conference on Acoustics, Speech and Signal Processing - ICASSP '07, Honolulu, HI, 2007, pp. IV-229-IV-232.

3.4.4.17. Zero one loss

The **`zero_one_loss`** function computes the sum or the average of the 0-1 classification loss (L_{0-1}) over n_{samples}. By default, the function normalizes over the sample. To get the sum of the L_{0-1}, set `normalize` to `False`.

In multilabel classification, the **`zero_one_loss`** scores a subset as one if its labels strictly match the predictions, and as a zero if there are any errors. By default, the function returns the percentage of imperfectly predicted subsets. To get the count of such subsets instead, set `normalize` to `False`.

If $\hat{y}_i$ is the predicted value of the i-th sample and y_i is the corresponding true value, then the 0-1 loss L_{0-1} is defined as:

$$L_{0-1}(y, \hat{y}) = \frac{1}{n_\text{samples}} \sum_{i=0}^{n_\text{samples}-1} 1(\hat{y}_i \not= y_i)$$

where $1(x)$ is the indicator function. The zero-one loss can also be computed as $\text{zero-one loss} = 1 - \text{accuracy}$.

```
>>> from sklearn.metrics import zero_one_loss
>>> y_pred = [1, 2, 3, 4]
>>> y_true = [2, 2, 3, 4]
>>> zero_one_loss(y_true, y_pred)
0.25
>>> zero_one_loss(y_true, y_pred, normalize=False)
1.0
```

In the multilabel case with binary label indicators, where the first label set [0,1] has an error:

```
>>> zero_one_loss(np.array([[0, 1], [1, 1]]), np.ones((2, 2)))
0.5

>>> zero_one_loss(np.array([[0, 1], [1, 1]]), np.ones((2, 2)),  normalize=False)
1.0
```

Examples

- See Recursive feature elimination with cross-validation for an example of zero one loss usage to perform recursive feature elimination with cross-validation.

3.4.4.18. Brier score loss

The **`brier_score_loss`** function computes the Brier score for binary and multiclass probabilistic predictions and is equivalent to the mean squared error. Quoting Wikipedia:

> "The Brier score is a strictly proper scoring rule that measures the accuracy of probabilistic predictions. [...] [It] is applicable to tasks in which predictions must assign probabilities to a set of mutually exclusive discrete outcomes or classes."

Let the true labels for a set of N data points be encoded as a 1-of-K binary indicator matrix Y, i.e., $y_{i,k} = 1$ if sample i has label k taken from a set of K labels. Let $\hat{P}$ be a matrix of probability estimates with elements $\hat{p}_{i,k} \approx \operatorname{Pr}(y_{i,k} = 1)$. Following the original definition by [Brier1950], the Brier score is given by:

$$BS(Y, \hat{P}) = \frac{1}{N}\sum_{i=0}^{N-1}\sum_{k=0}^{K-1}(y_{i,k} - \hat{p}_{i,k})^{2}$$

The Brier score lies in the interval $[0, 2]$ and the lower the value the better the probability estimates are (the mean squared difference is smaller). Actually, the Brier score is a strictly proper scoring rule, meaning that it achieves the best score only when the estimated probabilities equal the true ones.

Note that in the binary case, the Brier score is usually divided by two and ranges between $[0,1]$. For binary targets $y_i \in \{0, 1\}$ and probability estimates $\hat{p}_i \approx \operatorname{Pr}(y_i = 1)$ for the positive class, the Brier score is then equal to:

$$BS(y, \hat{p}) = \frac{1}{N} \sum_{i=0}^{N - 1}(y_i - \hat{p}_i)^2$$

The **`brier_score_loss`** function computes the Brier score given the ground-truth labels and predicted probabilities, as returned by an estimator's `predict_proba` method. The `scale_by_half` parameter controls which of the two above definitions to follow.

```
>>> import numpy as np
>>> from sklearn.metrics import brier_score_loss
>>> y_true = np.array([0, 1, 1, 0])
>>> y_true_categorical = np.array(["spam", "ham", "ham", "spam"])
>>> y_prob = np.array([0.1, 0.9, 0.8, 0.4])
>>> brier_score_loss(y_true, y_prob)
0.055
>>> brier_score_loss(y_true, 1 - y_prob, pos_label=0)
0.055
>>> brier_score_loss(y_true_categorical, y_prob, pos_label="ham")
0.055
>>> brier_score_loss(
...     ["eggs", "ham", "spam"],
...     [[0.8, 0.1, 0.1], [0.2, 0.7, 0.1], [0.2, 0.2, 0.6]],
...     labels=["eggs", "ham", "spam"],
... )
0.146
```

> **Note**
>
> As a strictly proper scoring rules for probabilistic predictions, the Brier score assesses calibration (reliability) and discriminative power (resolution) of a model, as well as the randomness of the data (uncertainty) at the same time. This follows from the well-known Brier score decomposition of Murphy [Murphy1973]. As it is not clear which term dominates, the score is of limited use for assessing calibration alone (unless one computes each term of the decomposition). A lower Brier loss, for instance, does not necessarily mean a better calibrated model, it could also mean a worse calibrated model with much more discriminatory power, e.g. using many more features.

Examples

- See Probability calibration of classifiers for an example of Brier score loss usage to perform probability calibration of classifiers.

References

[Brier1950] G. Brier (1950). "Verification of forecasts expressed in terms of probability". Monthly Weather Review 78(1), 1-3

[Murphy1973] Allan H. Murphy (1973). "A New Vector Partition of the Probability Score" Journal of Applied Meteorology and Climatology, 12(4), 595-600

3.4.4.19. Class likelihood ratios

The **`class_likelihood_ratios`** function computes the positive and negative likelihood ratios $LR_\pm$ for binary classes, which can be interpreted as the ratio of post-test to pre-test odds as explained below. As a consequence, this metric is invariant w.r.t. the class prevalence (the number of samples in the positive class divided by the total number of samples) and **can be extrapolated between populations regardless of any possible class imbalance.**

The $LR_\pm$ metrics are therefore very useful in settings where the data available to learn and evaluate a classifier is a study population with nearly balanced classes, such as a case-control study, while the target application, i.e. the general population, has very low prevalence.

The positive likelihood ratio LR_+ is the probability of a classifier to correctly predict that a sample belongs to the positive class divided by the probability of predicting the positive class for a sample belonging to the negative class:

$$LR_+ = \frac{\text{PR}(P+|T+)}{\text{PR}(P+|T-)}.$$

The notation here refers to predicted (P) or true (T) label and the sign $+$ and $-$ refer to the positive and negative class, respectively, e.g. $P+$ stands for "predicted positive".

Analogously, the negative likelihood ratio LR_- is the probability of a sample of the positive class being classified as belonging to the negative class divided by the probability of a sample of the negative class being correctly classified:

$$LR_- = \frac{\text{PR}(P-|T+)}{\text{PR}(P-|T-)}.$$

For classifiers above chance LR_+ above 1 **higher is better**, while LR_- ranges from 0 to 1 and **lower is better**. Values of $LR_\pm\approx 1$ correspond to chance level.

Notice that probabilities differ from counts, for instance $\operatorname{PR}(P+|T+)$ is not equal to the number of true positive counts `tp` (see the wikipedia page for the actual formulas).

Examples

- Class Likelihood Ratios to measure classification performance

Interpretation across varying prevalence

Mathematical divergences

References

3.4.4.20. D² score for classification

The D² score computes the fraction of deviance explained. It is a generalization of R², where the squared error is generalized and replaced by a classification deviance of choice $\text{dev}(y, \hat{y})$ (e.g., Log loss, Brier score,). D² is a form of a *skill score*. It is calculated as

$$D^2(y, \hat{y}) = 1 - \frac{\text{dev}(y, \hat{y})}{\text{dev}(y, y_{\text{null}})} \,.$$

Where y_{null} is the optimal prediction of an intercept-only model (e.g., the per-class proportion of `y_true` in the case of the Log loss and Brier score).

Like R², the best possible score is 1.0 and it can be negative (because the model can be arbitrarily worse). A constant model that always predicts y_{null}, disregarding the input features, would get a D² score of 0.0.

D2 log loss score

D2 Brier score

3.4.5. Multilabel ranking metrics

In multilabel learning, each sample can have any number of ground truth labels associated with it. The goal is to give high scores and better rank to the ground truth labels.

3.4.5.1. Coverage error

The **`coverage_error`** function computes the average number of labels that have to be included in the final prediction such that all true labels are predicted. This is useful if you want to know how many top-scored-labels you have to predict in average without missing any true one. The best value of this metric is thus the average number of true labels.

> **Note**
>
> Our implementation's score is 1 greater than the one given in Tsoumakas et al., 2010. This extends it to handle the degenerate case in which an instance has 0 true labels.

Formally, given a binary indicator matrix of the ground truth labels $y \in \left\{0, 1\right\}^{n_\text{samples} \times n_\text{labels}}$ and the score associated with each label $\hat{f} \in \mathbb{R}^{n_\text{samples} \times n_\text{labels}}$, the coverage is defined as

$$coverage(y, \hat{f}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}} - 1} \max_{j:y_{ij} = 1} \text{rank}_{ij}$$

with $\text{rank}_{ij} = \left|\left\{k: \hat{f}_{ik} \geq \hat{f}_{ij} \right\}\right|$. Given the rank definition, ties in `y_scores` are broken by giving the maximal rank that would have been assigned to all tied values.

Here is a small example of usage of this function:

```
>>> import numpy as np
>>> from sklearn.metrics import coverage_error
>>> y_true = np.array([[1, 0, 0], [0, 0, 1]])
>>> y_score = np.array([[0.75, 0.5, 1], [1, 0.2, 0.1]])
>>> coverage_error(y_true, y_score)
2.5
```

3.4.5.2. Label ranking average precision

The **`label_ranking_average_precision_score`** function implements label ranking average precision (LRAP). This metric is linked to the **`average_precision_score`** function, but is based on the notion of label ranking instead of precision and recall.

Label ranking average precision (LRAP) averages over the samples the answer to the following question: for each ground truth label, what fraction of higher-ranked labels were true labels? This performance measure will be higher if you are able to give better rank to the labels associated with each sample. The obtained score is always strictly greater than 0, and the best value is 1. If there is exactly one relevant label per sample, label ranking average precision is equivalent to the mean reciprocal rank.

Formally, given a binary indicator matrix of the ground truth labels $y \in \left\{0, 1\right\}^{n_\text{samples} \times n_\text{labels}}$ and the score associated with each label $\hat{f} \in \mathbb{R}^{n_\text{samples} \times n_\text{labels}}$, the average precision is defined as

$$LRAP(y, \hat{f}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}} - 1} \frac{1}{||y_i||_0} \sum_{j:y_{ij} = 1} \frac{|\mathcal{L}_{ij}|}{\text{rank}_{ij}}$$

where $\mathcal{L}_{ij} = \left\{k: y_{ik} = 1, \hat{f}_{ik} \geq \hat{f}_{ij} \right\}$, $\text{rank}_{ij} = \left|\left\{k: \hat{f}_{ik} \geq \hat{f}_{ij} \right\}\right|$, $|\cdot|$ computes the cardinality of the set (i.e., the number of elements in the set), and $||\cdot||_0$ is the ℓ_0 "norm" (which computes the number of nonzero elements in a vector).

Here is a small example of usage of this function:

```
>>> import numpy as np
>>> from sklearn.metrics import label_ranking_average_precision_score
>>> y_true = np.array([[1, 0, 0], [0, 0, 1]])
>>> y_score = np.array([[0.75, 0.5, 1], [1, 0.2, 0.1]])
>>> label_ranking_average_precision_score(y_true, y_score)
0.416
```

3.4.5.3. Ranking loss

The **`label_ranking_loss`** function computes the ranking loss which averages over the samples the number of label pairs that are incorrectly ordered, i.e. true labels have a lower score than false labels, weighted by the inverse of the number of ordered pairs of false and true labels. The lowest achievable ranking loss is zero.

Formally, given a binary indicator matrix of the ground truth labels $y \in \{0, 1\}^{n_\text{samples} \times n_\text{labels}}$ and the score associated with each label $\hat{f} \in \mathbb{R}^{n_\text{samples} \times n_\text{labels}}$, the ranking loss is defined as

$$ranking\_loss(y, \hat{f}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}} - 1} \frac{1}{||y_i||_0(n_\text{labels} - ||y_i||_0)} \left\{(k, l): \hat{f}_{ik} \leq \hat{f}_{il}, y_{ik} = 1, y_{il} = \right.$$

where $|\cdot|$ computes the cardinality of the set (i.e., the number of elements in the set) and $||\cdot||_0$ is the ℓ_0 "norm" (which computes the number of nonzero elements in a vector).

Here is a small example of usage of this function:

```
>>> import numpy as np
>>> from sklearn.metrics import label_ranking_loss
>>> y_true = np.array([[1, 0, 0], [0, 0, 1]])
>>> y_score = np.array([[0.75, 0.5, 1], [1, 0.2, 0.1]])
>>> label_ranking_loss(y_true, y_score)
0.75
>>> # With the following prediction, we have perfect and minimal loss
>>> y_score = np.array([[1.0, 0.1, 0.2], [0.1, 0.2, 0.9]])
>>> label_ranking_loss(y_true, y_score)
0.0
```

References

3.4.5.4. Normalized Discounted Cumulative Gain

Discounted Cumulative Gain (DCG) and Normalized Discounted Cumulative Gain (NDCG) are ranking metrics implemented in **`dcg_score`** and **`ndcg_score`** ; they compare a predicted order to ground-truth scores, such as the relevance of answers to a query.

From the Wikipedia page for Discounted Cumulative Gain:

"Discounted cumulative gain (DCG) is a measure of ranking quality. In information retrieval, it is often used to measure effectiveness of web search engine algorithms or related applications. Using a graded relevance scale of documents in a search-engine result set, DCG measures the usefulness, or gain, of a document based on its position in the result list. The gain is accumulated from the top of the result list to the bottom, with the gain of each result discounted at lower ranks."

DCG orders the true targets (e.g. relevance of query answers) in the predicted order, then multiplies them by a logarithmic decay and sums the result. The sum can be truncated after the first K results, in which case we call it DCG@K. NDCG, or NDCG@K is DCG divided by the DCG obtained by a perfect prediction, so that it is always between 0 and 1. Usually, NDCG is preferred to DCG.

Compared with the ranking loss, NDCG can take into account relevance scores, rather than a ground-truth ranking. So if the ground-truth consists only of an ordering, the ranking loss should be preferred; if the ground-truth consists of actual usefulness scores (e.g. 0 for irrelevant, 1 for relevant, 2 for very relevant), NDCG can be used.

For one sample, given the vector of continuous ground-truth values for each target $y \in \mathbb{R}^{M}$, where M is the number of outputs, and the prediction $\hat{y}$, which induces the ranking function f, the DCG score is

$$\sum_{r=1}^{\min(K, M)}\frac{y_{f(r)}}{\log(1 + r)}$$

and the NDCG score is the DCG score divided by the DCG score obtained for y.

References

3.4.6. Regression metrics

The **`sklearn.metrics`** module implements several loss, score, and utility functions to measure regression performance. Some of those have been enhanced to handle the multioutput case: **`mean_squared_error`**, **`mean_absolute_error`**, **`r2_score`**, **`explained_variance_score`**, **`mean_pinball_loss`**, **`d2_pinball_score`** and **`d2_absolute_error_score`**.

These functions have a `multioutput` keyword argument which specifies the way the scores or losses for each individual target should be averaged. The default is `'uniform_average'`, which specifies a uniformly weighted mean over outputs. If an `ndarray` of shape `(n_outputs,)` is passed, then its entries are interpreted as weights and an according weighted average is returned. If `multioutput` is `'raw_values'`, then all unaltered individual scores or losses will be returned in an array of shape `(n_outputs,)`.

The **`r2_score`** and **`explained_variance_score`** accept an additional value `'variance_weighted'` for the `multioutput` parameter. This option leads to a weighting of each individual score by the variance of the corresponding target variable. This setting quantifies the globally captured unscaled variance. If the target variables are of different scale, then this score puts more importance on explaining the higher variance variables.

3.4.6.1. R² score, the coefficient of determination

The **`r2_score`** function computes the coefficient of determination, usually denoted as R^2.

It represents the proportion of variance (of y) that has been explained by the independent variables in the model. It provides an indication of goodness of fit and therefore a measure of how well unseen samples are likely to be predicted by the model, through the proportion of explained variance.

As such variance is dataset dependent, R^2 may not be meaningfully comparable across different datasets. Best possible score is 1.0 and it can be negative (because the model can be arbitrarily worse). A constant model that always predicts the expected (average) value of y, disregarding the input features, would get an R^2 score of 0.0.

Note: when the prediction residuals have zero mean, the R^2 score and the Explained variance score are identical.

If $\hat{y}_i$ is the predicted value of the i-th sample and y_i is the corresponding true value for total n samples, the estimated R^2 is defined as:

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

where $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$ and $\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \epsilon_i^2$.

Note that **`r2_score`** calculates unadjusted R^2 without correcting for bias in sample variance of y.

In the particular case where the true target is constant, the R^2 score is not finite: it is either `NaN` (perfect predictions) or `-Inf` (imperfect predictions). Such non-finite scores may prevent correct model optimization such as grid-search cross-validation to be performed correctly. For this reason the default behaviour of **`r2_score`** is to replace them with 1.0 (perfect predictions) or 0.0 (imperfect predictions). If `force_finite` is set to `False`, this score falls back on the original R^2 definition.

Here is a small example of usage of the **`r2_score`** function:

```
>>> from sklearn.metrics import r2_score
>>> y_true = [3, -0.5, 2, 7]
>>> y_pred = [2.5, 0.0, 2, 8]
>>> r2_score(y_true, y_pred)
0.948
>>> y_true = [[0.5, 1], [-1, 1], [7, -6]]
>>> y_pred = [[0, 2], [-1, 2], [8, -5]]
>>> r2_score(y_true, y_pred, multioutput='variance_weighted')
0.938
>>> y_true = [[0.5, 1], [-1, 1], [7, -6]]
>>> y_pred = [[0, 2], [-1, 2], [8, -5]]
>>> r2_score(y_true, y_pred, multioutput='uniform_average')
0.936
>>> r2_score(y_true, y_pred, multioutput='raw_values')
array([0.965, 0.908])
>>> r2_score(y_true, y_pred, multioutput=[0.3, 0.7])
0.925
>>> y_true = [-2, -2, -2]
>>> y_pred = [-2, -2, -2]
>>> r2_score(y_true, y_pred)
1.0
>>> r2_score(y_true, y_pred, force_finite=False)
nan
>>> y_true = [-2, -2, -2]
>>> y_pred = [-2, -2, -2 + 1e-8]
>>> r2_score(y_true, y_pred)
0.0
>>> r2_score(y_true, y_pred, force_finite=False)
-inf
```

Examples

- See L1-based models for Sparse Signals for an example of R² score usage to evaluate Lasso and Elastic Net on sparse signals.

3.4.6.2. Mean absolute error

The **`mean_absolute_error`** function computes mean absolute error, a risk metric corresponding to the expected value of the absolute error loss or $l1$-norm loss.

If $\hat{y}_i$ is the predicted value of the i-th sample, and y_i is the corresponding true value, then the mean absolute error (MAE) estimated over n_{samples} is defined as

$$\text{MAE}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} \left| y_i - \hat{y}_i \right|.$$

Here is a small example of usage of the **`mean_absolute_error`** function:

```
>>> from sklearn.metrics import mean_absolute_error
>>> y_true = [3, -0.5, 2, 7]
>>> y_pred = [2.5, 0.0, 2, 8]
>>> mean_absolute_error(y_true, y_pred)
0.5
>>> y_true = [[0.5, 1], [-1, 1], [7, -6]]
>>> y_pred = [[0, 2], [-1, 2], [8, -5]]
>>> mean_absolute_error(y_true, y_pred)
0.75
>>> mean_absolute_error(y_true, y_pred, multioutput='raw_values')
array([0.5, 1. ])
>>> mean_absolute_error(y_true, y_pred, multioutput=[0.3, 0.7])
0.85
```

3.4.6.3. Mean squared error

The **`mean_squared_error`** function computes mean squared error, a risk metric corresponding to the expected value of the squared (quadratic) error or loss.

If $\hat{y}_i$ is the predicted value of the i-th sample, and y_i is the corresponding true value, then the mean squared error (MSE) estimated over n_{samples} is defined as

$$\text{MSE}(y, \hat{y}) = \frac{1}{n_\text{samples}} \sum_{i=0}^{n_\text{samples} - 1} (y_i - \hat{y}_i)^2.$$

Here is a small example of usage of the **`mean_squared_error`** function:

```
>>> from sklearn.metrics import mean_squared_error
>>> y_true = [3, -0.5, 2, 7]
>>> y_pred = [2.5, 0.0, 2, 8]
>>> mean_squared_error(y_true, y_pred)
0.375
>>> y_true = [[0.5, 1], [-1, 1], [7, -6]]
>>> y_pred = [[0, 2], [-1, 2], [8, -5]]
>>> mean_squared_error(y_true, y_pred)
0.7083
```

Examples

- See Gradient Boosting regression for an example of mean squared error usage to evaluate gradient boosting regression.

Taking the square root of the MSE, called the root mean squared error (RMSE), is another common metric that provides a measure in the same units as the target variable. RMSE is available through the **`root_mean_squared_error`** function.

3.4.6.4. Mean squared logarithmic error

The **`mean_squared_log_error`** function computes a risk metric corresponding to the expected value of the squared logarithmic (quadratic) error or loss.

If $\hat{y}_i$ is the predicted value of the i-th sample, and y_i is the corresponding true value, then the mean squared logarithmic error (MSLE) estimated over n_{samples} is defined as

$$\text{MSLE}(y, \hat{y}) = \frac{1}{n_\text{samples}} \sum_{i=0}^{n_\text{samples} - 1} (\log_e (1 + y_i) - \log_e (1 + \hat{y}_i))^2.$$

Where $\log_e (x)$ means the natural logarithm of x. This metric is best to use when targets having exponential growth, such as population counts, average sales of a commodity over a span of years etc. Note that this metric penalizes an under-predicted estimate greater than an over-predicted estimate.

Here is a small example of usage of the **`mean_squared_log_error`** function:

```
>>> from sklearn.metrics import mean_squared_log_error
>>> y_true = [3, 5, 2.5, 7]
>>> y_pred = [2.5, 5, 4, 8]
>>> mean_squared_log_error(y_true, y_pred)
0.0397
>>> y_true = [[0.5, 1], [1, 2], [7, 6]]
>>> y_pred = [[0.5, 2], [1, 2.5], [8, 8]]
>>> mean_squared_log_error(y_true, y_pred)
0.044
```

The root mean squared logarithmic error (RMSLE) is available through the **`root_mean_squared_log_error`** function.

3.4.6.5. Mean absolute percentage error

The **`mean_absolute_percentage_error`** (MAPE), also known as mean absolute percentage deviation (MAPD), is an evaluation metric for regression problems. The idea of this metric is to be sensitive to relative errors. It is for example not changed by a global scaling of the target variable.

If $\hat{y}_i$ is the predicted value of the i-th sample and y_i is the corresponding true value, then the mean absolute percentage error (MAPE) estimated over n_{samples} is defined as

$$\text{MAPE}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} \frac{{}\left| y_i - \hat{y}_i \right|}{\max(\epsilon, \left| y_i \right|)}$$

where ϵ is an arbitrary small yet strictly positive number to avoid undefined results when y is zero.

The **`mean_absolute_percentage_error`** function supports multioutput.

Here is a small example of usage of the **`mean_absolute_percentage_error`** function:

```
>>> from sklearn.metrics import mean_absolute_percentage_error
>>> y_true = [1, 10, 1e6]
>>> y_pred = [0.9, 15, 1.2e6]
>>> mean_absolute_percentage_error(y_true, y_pred)
0.2666
```

In above example, if we had used `mean_absolute_error`, it would have ignored the small magnitude values and only reflected the error in prediction of highest magnitude value. But that problem is resolved in case of MAPE because it calculates relative percentage error with respect to actual output.

> **Note**
>
> The MAPE formula here does not represent the common "percentage" definition: the percentage in the range [0, 100] is converted to a relative value in the range [0, 1] by dividing by 100. Thus, an error of 200% corresponds to a relative error of 2. The motivation here is to have a range of values that is more consistent with other error metrics in scikit-learn, such as `accuracy_score`.
>
> To obtain the mean absolute percentage error as per the Wikipedia formula, multiply the `mean_absolute_percentage_error` computed here by 100.

References

3.4.6.6. Median absolute error

The **`median_absolute_error`** is particularly interesting because it is robust to outliers. The loss is calculated by taking the median of all absolute differences between the target and the prediction.

If $\hat{y}_i$ is the predicted value of the i-th sample and y_i is the corresponding true value, then the median absolute error (MedAE) estimated over n_{samples} is defined as

$$\text{MedAE}(y, \hat{y}) = \text{median}(\mid y_1 - \hat{y}_1 \mid, \ldots, \mid y_n - \hat{y}_n \mid).$$

The **`median_absolute_error`** does not support multioutput.

Here is a small example of usage of the **`median_absolute_error`** function:

```
>>> from sklearn.metrics import median_absolute_error
>>> y_true = [3, -0.5, 2, 7]
>>> y_pred = [2.5, 0.0, 2, 8]
>>> median_absolute_error(y_true, y_pred)
0.5
```

3.4.6.7. Max error

The **`max_error`** function computes the maximum residual error , a metric that captures the worst case error between the predicted value and the true value. In a perfectly fitted single output regression model, `max_error` would be `0` on the training set and though this would be highly unlikely in the real world, this metric shows the extent of error that the model had when it was fitted.

If $\hat{y}_i$ is the predicted value of the i-th sample, and y_i is the corresponding true value, then the max error is defined as

$$\text{Max Error}(y, \hat{y}) = \max(| y_i - \hat{y}_i |)$$

Here is a small example of usage of the **`max_error`** function:

```
>>> from sklearn.metrics import max_error
>>> y_true = [3, 2, 7, 1]
>>> y_pred = [9, 2, 7, 1]
>>> max_error(y_true, y_pred)
6.0
```

The **`max_error`** does not support multioutput.

3.4.6.8. Explained variance score

The **`explained_variance_score`** computes the explained variance regression score.

If $\hat{y}$ is the estimated target output, y the corresponding (correct) target output, and Var is Variance, the square of the standard deviation, then the explained variance is estimated as follow:

$$explained\_{}variance(y, \hat{y}) = 1 - \frac{Var\{ y - \hat{y}\}}{Var\{y\}}$$

The best possible score is 1.0, lower values are worse.

> **Link to R² score, the coefficient of determination**
>
> The difference between the explained variance score and the R² score, the coefficient of determination is that the explained variance score does not account for systematic offset in the prediction. For this reason, the R² score, the coefficient of determination should be preferred in general.

In the particular case where the true target is constant, the Explained Variance score is not finite: it is either `NaN` (perfect predictions) or `-Inf` (imperfect predictions). Such non-finite scores may prevent correct model optimization such as grid-search cross-validation to be performed correctly. For this reason the default behaviour of **`explained_variance_score`** is to replace them with 1.0 (perfect predictions) or 0.0 (imperfect predictions). You can set the `force_finite` parameter to `False` to prevent this fix from happening and fallback on the original Explained Variance score.

Here is a small example of usage of the **`explained_variance_score`** function:

```
>>> from sklearn.metrics import explained_variance_score
>>> y_true = [3, -0.5, 2, 7]
>>> y_pred = [2.5, 0.0, 2, 8]
>>> explained_variance_score(y_true, y_pred)
0.957
>>> y_true = [[0.5, 1], [-1, 1], [7, -6]]
>>> y_pred = [[0, 2], [-1, 2], [8, -5]]
>>> explained_variance_score(y_true, y_pred, multioutput='raw_values')
array([0.967, 1.        ])
>>> explained_variance_score(y_true, y_pred, multioutput=[0.3, 0.7])
0.990
>>> y_true = [-2, -2, -2]
>>> y_pred = [-2, -2, -2]
>>> explained_variance_score(y_true, y_pred)
1.0
>>> explained_variance_score(y_true, y_pred, force_finite=False)
nan
>>> y_true = [-2, -2, -2]
>>> y_pred = [-2, -2, -2 + 1e-8]
>>> explained_variance_score(y_true, y_pred)
0.0
>>> explained_variance_score(y_true, y_pred, force_finite=False)
-inf
```

3.4.6.9. Mean Poisson, Gamma, and Tweedie deviances

The **`mean_tweedie_deviance`** function computes the mean Tweedie deviance error with a `power` parameter (p). This is a metric that elicits predicted expectation values of regression targets.

Following special cases exist,

- when `power=0` it is equivalent to **`mean_squared_error`**.
- when `power=1` it is equivalent to **`mean_poisson_deviance`**.
- when `power=2` it is equivalent to **`mean_gamma_deviance`**.

If $\hat{y}_i$ is the predicted value of the i-th sample, and y_i is the corresponding true value, then the mean Tweedie deviance error (D) for power p, estimated over n_{samples} is defined as

$$\text{D}(y, \hat{y}) = \frac{1}{n_\text{samples}} \sum_{i=0}^{n_\text{samples} - 1} \begin{cases} (y_i-\hat{y}_i)^2, & \text{for }p=0\text{ (Normal)}\\ 2(y_i \log(y_i/\hat{y}_i) + \hat{y}_i - y_i), & \text{for }p=1\text{ (Poisson)}\\ 2(\log(\hat{y}_i/y_i) + y_i/\hat{y}_i - 1), & \text{for }p=2\text{ (Gamma)}\\ 2\left(\frac{\max(y_i,0)^{2-p}}{(1-p)(2-p)}- \frac{y_i\,\hat{y}_i^{1-p}}{1-p}+\frac{\hat{y}_i^{2-p}}{2-p}\right), & \text{otherwise} \end{cases}$$

Tweedie deviance is a homogeneous function of degree `2-power`. Thus, Gamma distribution with `power=2` means that simultaneously scaling `y_true` and `y_pred` has no effect on the deviance. For Poisson distribution `power=1` the deviance scales linearly, and for Normal distribution (`power=0`), quadratically. In general, the higher `power` the less weight is given to extreme deviations between true and predicted targets.

For instance, let's compare the two predictions 1.5 and 150 that are both 50% larger than their corresponding true value.

The mean squared error (`power=0`) is very sensitive to the prediction difference of the second point,:

```
>>> from sklearn.metrics import mean_tweedie_deviance
>>> mean_tweedie_deviance([1.0], [1.5], power=0)
0.25
>>> mean_tweedie_deviance([100.], [150.], power=0)
2500.0
```

If we increase `power` to 1,:

```
>>> mean_tweedie_deviance([1.0], [1.5], power=1)
0.189
>>> mean_tweedie_deviance([100.], [150.], power=1)
18.9
```

the difference in errors decreases. Finally, by setting, `power=2`:

```
>>> mean_tweedie_deviance([1.0], [1.5], power=2)
0.144
>>> mean_tweedie_deviance([100.], [150.], power=2)
0.144
```

we would get identical errors. The deviance when `power=2` is thus only sensitive to relative errors.

3.4.6.10. Pinball loss

The **`mean_pinball_loss`** function is used to evaluate the predictive performance of quantile regression models.

$$\text{pinball}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} \alpha \max(y_i - \hat{y}_i, 0) + (1 - \alpha) \max(\hat{y}_i - y_i, 0)$$

The value of pinball loss is equivalent to half of **`mean_absolute_error`** when the quantile parameter `alpha` is set to 0.5.

Here is a small example of usage of the **`mean_pinball_loss`** function:

```
>>> from sklearn.metrics import mean_pinball_loss
>>> y_true = [1, 2, 3]
>>> mean_pinball_loss(y_true, [0, 2, 3], alpha=0.1)
0.033
>>> mean_pinball_loss(y_true, [1, 2, 4], alpha=0.1)
0.3
>>> mean_pinball_loss(y_true, [0, 2, 3], alpha=0.9)
0.3
>>> mean_pinball_loss(y_true, [1, 2, 4], alpha=0.9)
0.033
>>> mean_pinball_loss(y_true, y_true, alpha=0.1)
0.0
>>> mean_pinball_loss(y_true, y_true, alpha=0.9)
0.0
```

It is possible to build a scorer object with a specific choice of `alpha`:

```
>>> from sklearn.metrics import make_scorer
>>> mean_pinball_loss_95p = make_scorer(mean_pinball_loss, alpha=0.95)
```

Such a scorer can be used to evaluate the generalization performance of a quantile regressor via cross-validation:

```
>>> from sklearn.datasets import make_regression
>>> from sklearn.model_selection import cross_val_score
>>> from sklearn.ensemble import GradientBoostingRegressor
>>>
>>> X, y = make_regression(n_samples=100, random_state=0)
>>> estimator = GradientBoostingRegressor(
...     loss="quantile",
...     alpha=0.95,
...     random_state=0,
... )
>>> cross_val_score(estimator, X, y, cv=5, scoring=mean_pinball_loss_95p)
array([14.3,  9.8, 23.9,  9.4, 10.8])
```

It is also possible to build scorer objects for hyper-parameter tuning. The sign of the loss must be switched to ensure that greater means better as explained in the example linked below.

Examples

- See Prediction Intervals for Gradient Boosting Regression for an example of using the pinball loss to evaluate and tune the hyper-parameters of quantile regression models on data with non-symmetric noise and outliers.

3.4.6.11. D² score

The D² score computes the fraction of deviance explained. It is a generalization of R², where the squared error is generalized and replaced by a deviance of choice $\text{dev}(y, \hat{y})$ (e.g., Tweedie, pinball or mean absolute error). D² is a form of a *skill score*. It is calculated as

$$D^2(y, \hat{y}) = 1 - \frac{\text{dev}(y, \hat{y})}{\text{dev}(y, y_{\text{null}})} \,.$$

Where y_{null} is the optimal prediction of an intercept-only model (e.g., the mean of `y_true` for the Tweedie case, the median for absolute error and the alpha-quantile for pinball loss).

Like R², the best possible score is 1.0 and it can be negative (because the model can be arbitrarily worse). A constant model that always predicts y_{null}, disregarding the input features, would get a D² score of 0.0.

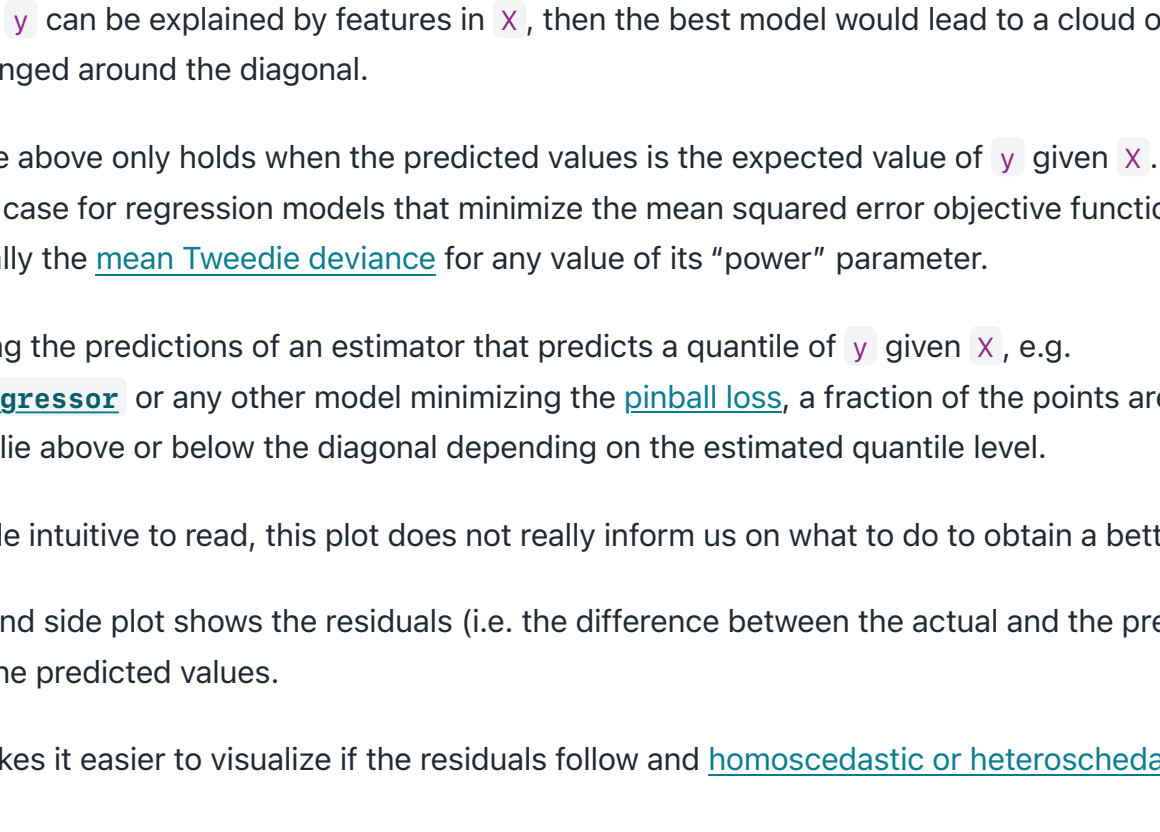
D² Tweedie score

D² pinball score

D² absolute error score

3.4.6.12. Visual evaluation of regression models

Among methods to assess the quality of regression models, scikit-learn provides the **`PredictionErrorDisplay`** class. It allows to visually inspect the prediction errors of a model in two different manners.



The plot on the left shows the actual values vs predicted values. For a noise-free regression task aiming to predict the (conditional) expectation of `y`, a perfect regression model would display data points on the diagonal defined by predicted equal to actual values. The further away from this optimal line, the larger the error of the model. In a more realistic setting with irreducible noise, that is, when not all the variations of `y` can be explained by features in `X`, then the best model would lead to a cloud of points densely arranged around the diagonal.

Note that the above only holds when the predicted values is the expected value of `y` given `X`. This is typically the case for regression models that minimize the mean squared error objective function or more generally the mean Tweedie deviance for any value of its "power" parameter.

When plotting the predictions of an estimator that predicts a quantile of `y` given `X`, e.g. **`QuantileRegressor`** or any other model minimizing the pinball loss, a fraction of the points are either expected to lie above or below the diagonal depending on the estimated quantile level.

All in all, while intuitive to read, this plot does not really inform us on what to do to obtain a better model.

The right-hand side plot shows the residuals (i.e. the difference between the actual and the predicted values) vs. the predicted values.

This plot makes it easier to visualize if the residuals follow and homoscedastic or heteroschedastic distribution.

In particular, if the true distribution of `y|X` is Poisson or Gamma distributed, it is expected that the variance of the residuals of the optimal model would grow with the predicted value of `E[y|X]` (either linearly for Poisson or quadratically for Gamma).

When fitting a linear least squares regression model (see **`LinearRegression`** and **`Ridge`**), we can use this plot to check if some of the model assumptions are met, in particular that the residuals should be uncorrelated, their expected value should be null and that their variance should be constant (homoschedasticity).

If this is not the case, and in particular if the residuals plot show some banana-shaped structure, this is a hint that the model is likely misspecified and that non-linear feature engineering or switching to a non-linear regression model might be useful.

Refer to the example below to see a model evaluation that makes use of this display.

Examples

- See Effect of transforming the targets in regression model for an example on how to use **`PredictionErrorDisplay`** to visualize the prediction quality improvement of a regression model obtained by transforming the target before learning.

3.4.7. Clustering metrics

The **`sklearn.metrics`** module implements several loss, score, and utility functions to measure clustering performance. For more information see the Clustering performance evaluation section for instance clustering, and Biclustering evaluation for biclustering.

3.4.8. Dummy estimators

When doing supervised learning, a simple sanity check consists of comparing one's estimator against simple rules of thumb. **`DummyClassifier`** implements several such simple strategies for classification:

- `stratified` generates random predictions by respecting the training set class distribution.
- `most_frequent` always predicts the most frequent label in the training set.
- `prior` always predicts the class that maximizes the class prior (like `most_frequent`) and `predict_proba` returns the class prior.
- `uniform` generates predictions uniformly at random.
- **`constant` always predicts a constant label that is provided by the user.**
    - A major motivation of this method is F1-scoring, when the positive class is in the minority.

A major motivation of this method is F1-scoring, when the positive class is in the minority.

Note that with all these strategies, the `predict` method completely ignores the input data!

To illustrate **`DummyClassifier`**, first let's create an imbalanced dataset:

```
>>> from sklearn.datasets import load_iris
>>> from sklearn.model_selection import train_test_split
>>> X, y = load_iris(return_X_y=True)
>>> y[y != 1] = -1
>>> X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=0)
```

Next, let's compare the accuracy of `SVC` and `most_frequent`:

```
>>> from sklearn.dummy import DummyClassifier
>>> from sklearn.svm import SVC
>>> clf = SVC(kernel='linear', C=1).fit(X_train, y_train)
>>> clf.score(X_test, y_test)
0.63
>>> clf = DummyClassifier(strategy='most_frequent', random_state=0)
>>> clf.fit(X_train, y_train)
DummyClassifier(random_state=0, strategy='most_frequent')
>>> clf.score(X_test, y_test)
0.579
```

We see that `SVC` doesn't do much better than a dummy classifier. Now, let's change the kernel:

```
>>> clf = SVC(kernel='rbf', C=1).fit(X_train, y_train)
>>> clf.score(X_test, y_test)
0.94
```

We see that the accuracy was boosted to almost 100%. A cross validation strategy is recommended for a better estimate of the accuracy, if it is not too CPU costly. For more information see the Cross-validation: evaluating estimator performance section. Moreover if you want to optimize over the parameter space, it is highly recommended to use an appropriate methodology; see the Tuning the hyper-parameters of an estimator section for details.

More generally, when the accuracy of a classifier is too close to random, it probably means that something went wrong: features are not helpful, a hyperparameter is not correctly tuned, the classifier is suffering from class imbalance, etc…

`DummyRegressor` also implements four simple rules of thumb for regression:

- `mean` always predicts the mean of the training targets.
- `median` always predicts the median of the training targets.
- `quantile` always predicts a user provided quantile of the training targets.
- `constant` always predicts a constant value that is provided by the user.

In all these strategies, the `predict` method completely ignores the input data.

Previous
3.3. Tuning the decision threshold for class prediction

Next
3.5. Validation curves: plotting scores to evaluate models

© Copyright 2007 - 2026, scikit-learn developers (BSD License).